From Noisy Data to a Distribution over Equations

The algorithm behind the pipeline: spectral core, Bayesian layers, grouping and the loop

Explained edition

October 2026

Abstract

The analytic theory of the project says what a single exact solution of a linear constant-coefficient equation $P(\partial_t, \partial_x)u = 0$ remembers about P: its factors, the components they generate, the interaction between terms. This document explains the *algorithm* that turns that theory into a procedure working on *noisy, scattered measurements*. Five layers are described. (i) *Interpolation*: a Gaussian-process posterior returns many plausible fields $\hat{u}_s$ with weights π_s. (ii) *Spectral core*, the only deterministic layer: on each $\hat{u}_s$ a Fourier transform in x and a matrix pencil in t produce points $(\lambda, i\kappa)$ of the dispersion variety, to which a *structured* family of factors is fitted and compared by BIC. (iii) *Aggregation*: the factors of all samples form a weighted cloud, whose Dirichlet-process mixture gives a posterior on the number of factors. (iv) *Grouping*, which acts in mode space and on rectangles of the domain, never on raw data points. (v) *A Bayesian output layer*, built on the Friedrichs angle, giving credible intervals on the weights and the probability that the split exists. A sixth part, the *loop* with its *refinement on the raw data*, decides and labels the final answer; a long series of fixes made it robust (a section is devoted to them). A set of optional *learned layers* (Phase 4) was built and evaluated, and mostly not adopted; it is told in the companion document. In the final benchmark the pipeline recovers the right factor set on all 30 inputs without a false-confident answer. The text says, for each part, what is implemented and tested, what is prototype, what is only planned, and what the theory proves impossible.

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About this explained edition

The structure follows the explained edition of the analytic paper. Every section opens with a box *In this section* (starting point, reason, role). Theorems and algorithm statements sit in blue boxes; **purple "Explained" boxes** read the surrounding material step by step with small examples; orange boxes give intuition; red boxes are warnings. Status labels are [implemented, tested] (code and tests exist), [implemented, uncommitted] (code exists, reported by its session, not yet committed), [prototype] (an experiment script only), [planned], and [open]. Grey tags such as (theory T5) point to rows of the file `analysis/THEORY_TO_ALGO.md`, which links each algorithmic choice to a result of the analytic work; Appendix 19 states each tagged result in one sentence. A new symbol is always explained where it first appears, and each section opens with a box "Symbols used here: what they are and where they come from."

1 Introduction

In this section

Starting from	noisy measurements (t_i, x_i, y_i) with $y_i \approx u(t_i, x_i)$ of a field that solves an unknown linear constant-coefficient equation in one time and one space variable; the analytic theory (identification, factor decomposition, interaction term) is taken as given.
Why this section	to turn exact theorems into a procedure whose input is imperfect data, and to say exactly where the exactness is lost.
Role in the paper	this section frames the three tasks, shows the whole pipeline in one figure and lists what each stage relies on. Sections 3–8 then describe the stages one by one.

1.1 The question

Throughout, $u(t, x)$ is a field (a function of time t and one space coordinate x) known on a rectangle Ω of the (t, x)-plane, and $P(\partial_t, \partial_x)u = 0$ is the linear constant-coefficient equation it solves, with P unknown; all the symbols are introduced in Section 2. The analytic work answers three questions for an exact solution u: which operator P does u determine (Q1), how does u split into the components generated by the factors of P (Q2), and what part of u is due to no single term but to their interplay (Q3). Here the data are finitely many noisy samples, so the answers must change form:

- **A1. Which factors, with which coefficients, and how sure are we?** The output is a *probability distribution over factor structures*, never a single "identified" law.
- **A2. How much of the signal does each factor carry?** Weights w_j with credible intervals that widen when the components are hard to separate.
- **A3. Are the factors really separate or tangled?** The numbers γ_N, c_F and the interaction ratio ρ, as distributions over interpolants.

The setting is no longer purely analytic: part of the algorithm is statistical. The design rule is that the delicate analytic part stays deterministic and is called as a black box, and everything around it is Bayesian.

1.2 The pipeline

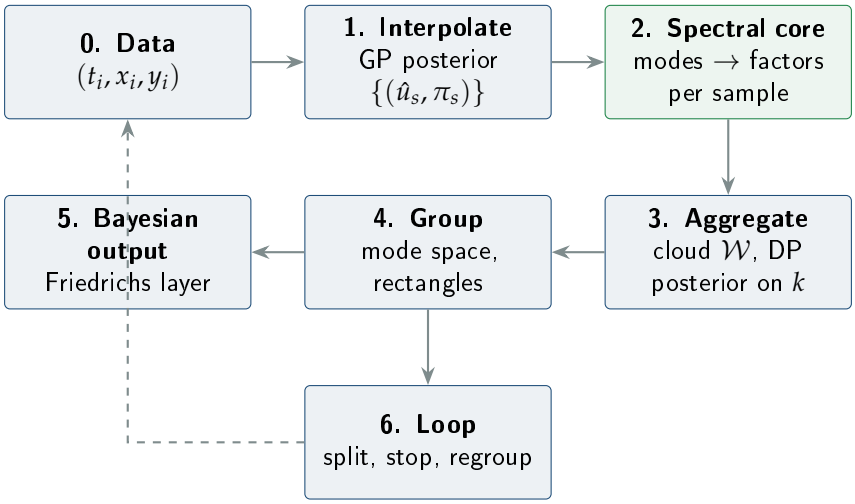


Figure 1: The pipeline. Only the green box is deterministic. The dashed arrow is the recursion: each group becomes a new data set.

1.3 Notation at a glance

Every symbol is introduced in Section 2 or in the box that opens the section that needs it; this table only collects the main ones and says where each comes from.

1.4 Roadmap

Section 2 recalls why exponentials are the right "basis". Section 3 is stage 1, Section 4 stage 2 (the longest), Section 5 stage 3, Section 6 stage 4, Section 7 stage 5 and Section 8 the loop. Section 10 answers the two design questions put to the machine-learning side, Section 11 describes the trainable version, Section 12 follows one example from data to output, and Section 14 lists the limits.

2 Notation from scratch: symbols, exponentials, the dispersion variety

In this section	
Starting from	nothing beyond calculus: a function $u(t, x)$ of time and one space variable, and derivatives. Every symbol used later is introduced here, in the order in which it is needed, with the reason it exists.
Why this section	the algorithm works on exponential modes, not on a chosen dictionary of functions. To follow it one must know what a symbol, a factor, a wavenumber and a mode are; this section builds them one at a time.
Role in the paper	foundation for everything after it. Each later section begins with a box listing the symbols it uses and where each comes from.

Stage	What it does	Relies on	Status
1 Interpolation	GP posterior on a grid from scattered data; hyperparameters marginalised	(theory T11), (theory T12)	[implemented, tested]; band coverage within ± 5 points in 11 of 12 cells
2 Spectral core	FFT in x, matrix pencil in t; structured factor fit, BIC over 85 structures; components, weights, γ_N; per-bin correlated noise	(theory T1), (theory T2), (theory T5), (theory T15)	[implemented, tested]
3 Aggregation	weighted cloud of factor coefficients; DP mixture; k_{eff}	(theory T12)	[implemented, tested]
4 Grouping	mode space, rectangles with quadtree	(theory T1), (theory T5), (theory T8)	[implemented, tested]
5 Bayesian output	posterior on components, $P(\text{split})$, intervals following θ_F; raw-data evidence	(theory T5), (theory T6)	[implemented, tested]; coverage logged 0.85–1.00, not certified
6 Loop	degree growth with a held-out guard, regroup at fixed k, recursion per factor, refinement on the observations	(theory T15)	[implemented, tested]; thresholds still hand defaults
6b Refinement and decision	raw-data evidence, support path, exact splits, goodness-of-fit downgrades, intervals (Sec. 9)	(theory T17), T44–T103	[implemented, tested]; final benchmark 30/30 right
7 Learned layers	seeds for the exponential fit, calibration, failure validator, remedy table (Phase 4; see the companion document)	(theory T10)	evaluated; none ships except an exact memo; pipeline = pipeline + ML

Table 1: The stages, the theory each one rests on, and their present status.

2.1 The objects: field, region, equation

Symbols used here: what they are and where they come from

t, x	time and one space coordinate (a rod, a string, a line)	the physical setting
Ω	the region of the (t, x)-plane where the field is known, a rectangle $[t_0, t_1] \times [x_0, x_1]$ here	the data
$u(t, x)$	the *field*: a number (real or complex) at each point of Ω (a temperature, a displacement)	the data, up to noise
∂_t, ∂_x	partial derivatives $\partial/\partial t$ and $\partial/\partial x$	calculus
$P(\partial_t, \partial_x)$	a *linear operator with constant coefficients*: a polynomial in the derivatives, e.g. $\partial_t - D\partial_x^2$ (heat); the equation is $Pu = 0$	the unknown of the problem

The problem is: given measurements of u on Ω, find P. "Linear with constant coefficients" means P is a sum of terms $c\,\partial_t^a \partial_x^b$ with numbers c (not functions of t or x), so that for example

Symbol	Meaning	Comes from
$u(t,x)$, Ω	the field, and the rectangle of the (t,x)-plane where it is known	the data; Sec. 2
P, Q_j	the unknown operator (as a polynomial in τ,ξ) and its factors	the theory; Sec. 2
κ, λ	wavenumber (spatial frequency, real) and rate of evolution in time (complex) of a mode $e^{\lambda t+i\kappa x}$	Sec. 2
m_j, dx_j, $\mathcal{M}$	degree in τ and bound on the degree in ξ of factor j; a *structure*, the list of these shapes (85 candidates)	Sec. 2; the algorithm
$\hat{u}_s$, π_s, N_s	the s-th posterior field, its weight, the number of fields	Sec. 3
σ_λ	uncertainty of a mode's rate λ (Cramér–Rao)	Sec. 4
u_j, w_j	component of factor j; its weight $\|u_j\| / \sum_k \|u_k\|$	Sec. 4
γ_N, c_F	how well the components can be separated; $\gamma_2 = 1 - c_F$	theory; Sec. 4
ρ	interaction ratio $\|E_{\text{int}}\| / \|u(T) - u_0\|$ of an evolution factor	theory; Sec. 4
$\boldsymbol{\beta}_j^{(s)}$	coefficient vector of factor j in sample s	Sec. 5
$\mathcal{W}$	the weighted cloud of all the $\boldsymbol{\beta}_j^{(s)}$	Sec. 5
k_{eff}	$\exp(-\sum_j W_j \log W_j)$, effective number of factors	Sec. 5

Table 2: Notation guide.

$\partial_t u + \partial_x u = 0$ (transport at speed 1) and $\partial_t u - D\,\partial_x^2 u = 0$ (heat, with diffusion constant $D > 0$) belong to the class, while $\partial_t u + u\,\partial_x u = 0$ (Burgers) does not.

2.2 Derivatives become polynomials: the symbol

Because the coefficients are constant, an operator is *multiplication-like*: composing $\partial_t^a \partial_x^b$ with $\partial_t^{a'} \partial_x^{b'}$ simply adds exponents, exactly as in a ring of polynomials. So one introduces two formal variables, τ standing for ∂_t and ξ standing for ∂_x, and writes the operator as a polynomial $P(\tau,\xi)$, its *symbol*. Example: $\partial_t - D\partial_x^2 \;\leftrightarrow\; \tau - D\xi^2$. The reason for the change of notation is that polynomials can be *factored*, and factoring is the central operation of the method: if $P(\tau,\xi) = Q_1(\tau,\xi)\,Q_2(\tau,\xi)$ then the operator is the composition of the two simpler operators $Q_1(\partial)$ and $Q_2(\partial)$. The Q_j are called the *factors* (they are the "basic behaviours" of the equation: transport, diffusion, …).

2.3 An exponential is an eigenfunction

For a field of the form $e^{\lambda t+\mu x}$ (where λ,μ are complex numbers) every derivative just multiplies by a number: $\partial_t e^{\lambda t+\mu x} = \lambda e^{\lambda t+\mu x}$ and $\partial_x e^{\lambda t+\mu x} = \mu\, e^{\lambda t+\mu x}$. Therefore

$$P(\partial_t, \partial_x)\, e^{\lambda t+\mu x} = P(\lambda,\mu)\, e^{\lambda t+\mu x}. \tag{1}$$

So $e^{\lambda t+\mu x}$ is a solution of $Pu = 0$ exactly when $P(\lambda,\mu) = 0$. The set $V(P) = \{(\lambda,\mu) \in \mathbb{C}^2 : P(\lambda,\mu) = 0\}$ is the *dispersion variety* (a curve in the plane of the two numbers); the word "dis-

persion" comes from physics, where λ as a function of the spatial frequency is called the dispersion relation.

2.4 The wavenumber κ and the rate λ

On a bounded region one expands in spatial exponentials that stay bounded in x, so μ is taken purely imaginary, $\mu = i\kappa$ with κ *real*. The two numbers then have clear roles:

- $\kappa \in \mathbb{R}$, the *wavenumber* (Fourier variable conjugate to x): which spatial exponential $e^{i\kappa x}$ we look at; its spatial period is $2\pi/|\kappa|$;
- $\lambda \in \mathbb{C}$, the *rate*: how the equation makes that exponential evolve in time; $\operatorname{Re}\lambda < 0$ means decay, $\operatorname{Im}\lambda \neq 0$ means oscillation or motion.

A pair (κ, λ) with $P(\lambda, i\kappa) = 0$ is a *mode*: the function $e^{\lambda t + i\kappa x}$ is an exact solution. Substituting $\mu = i\kappa$ in the symbol means $\xi = i\kappa$.

What κ is (and what it is not)

κ is the *Fourier variable* conjugate to x: the frequency of the exponential $e^{i\kappa x}$ in which a field is expanded. It is called a "wavenumber" only by convention, and it has *nothing to do with the wave equation*: it appears identically for transport, heat, beam, Schrödinger and so on. (If one wants a length, $2\pi/|\kappa|$ is the spatial period of that exponential.) The exponentials $e^{\lambda t + i\kappa x}$ are the "basis functions" of the method: κ labels which basis function, and λ is the rate at which the equation makes it evolve in time. The wave equation is just one of ten test cases.

2.5 The Fourier transform in x and the amplitudes

Because a general field is a superposition of modes, one separates them by the Fourier transform in x,

$$\hat{u}(t,\kappa) = \int u(t,x)\, e^{-i\kappa x}\, \mathrm{d}x \qquad \text{(on a grid: the discrete FFT)}.$$

If $u = \sum_m A_m\, e^{\lambda_m t + i\kappa_m x}$, then $\hat{u}(t,\kappa)$ keeps, at the frequency κ, only the modes with $\kappa_m = \kappa$: $\hat{u}(t,\kappa) = \sum_{m:\kappa_m=\kappa} A_m e^{\lambda_m t}$. The numbers $A_m \in \mathbb{C}$ are the *amplitudes* (how much of each mode is present). Extracting the λ_m and A_m from the time series $\hat{u}(\cdot,\kappa)$ is the job of Section 4.1.

Explained: from an equation to a curve of points

Step 1: transport. $u_t + u_x = 0$ has symbol $\tau + \xi$. A mode $e^{\lambda t + i\kappa x}$ is a solution iff $\lambda + i\kappa = 0$, i.e. $\lambda = -i\kappa$. In the (κ, λ) plane the modes lie on a straight line: a bump $g(x-t) = \int \hat{g}(\kappa) e^{i\kappa(x-t)}\, \mathrm{d}\kappa$ is a sum of modes with $\lambda = -i\kappa$.

Step 2: heat. $u_t = Du_{xx}$ has symbol $\tau - D\xi^2$, so $\lambda = D(i\kappa)^2 = -D\kappa^2$: a parabola, real and negative (every mode decays).

Step 3: wave. $u_{tt} = c^2 u_{xx}$ has symbol $\tau^2 - c^2\xi^2$, so $\lambda = \pm ic\kappa$: two lines through the origin. At $\kappa = 0$ the two roots coincide: this is the shared mode responsible for the ambiguity $\delta(P) = 1$ of the theory (explained in Section 4).

Step 4: the algorithm reads the curve, not the function. A solution of $P = (\tau + \xi)(\tau - D\xi^2)$ (transport plus a diffusing wave) produces, at each κ, *two* rates: $-i\kappa$ and $-D\kappa^2$. The set of all measured (κ, λ) sits on the union of a line and a parabola. Finding the factors of P means finding the curves on which the points lie: this is a geometric fitting problem, which is the content of Section 4.

2.6 Why exponentials are the right "basis"

The Ehrenpreis–Palamodov principle (a theorem of the analytic theory, (theory T25)) says that on a convex region every solution of $Pu = 0$ is a superposition (an integral) of such exponentials over $V(P)$. So in this class the exponentials are universal: no dictionary of functions has to be chosen. This is the reason the first design question (Section 10) has a clean answer.

2.7 Monic symbols, degrees, and structures

Symbols used here: what they are and where they come from

$Q(\tau, \xi)$	one factor of the symbol	factorisation of P
m	degree of Q in τ ("order in time"): $m = 1$ for transport, heat; $m = 2$ for wave, beam	the factor itself
$p_k(\xi)$	coefficients of Q as a polynomial in τ; each is a polynomial in ξ	the factor itself
dx	upper bound for the degrees of the p_k in ξ ("order in space"); a *bound*, not the true degree	chosen by the algorithm
structure $\mathcal{M}$	the list $((m_1, dx_1), \ldots, (m_k, dx_k))$ of shapes of the factors	chosen by the algorithm

The code fixes the shape of a factor to be *monic in* τ, meaning that its leading coefficient in τ is 1:

$$Q(\tau, \xi) = \tau^m + \sum_{k<m} p_k(\xi)\, \tau^k, \qquad \deg p_k \le dx. \tag{2}$$

Monic in τ is the setting in which the identification theorems apply (theory T13): the evolution in time determines everything (one can solve for $\partial_t^m u$). For example $\tau - a_1\xi$ is a factor with $m = 1$, $dx = 1$ (transport, speed $-a_1$), $\tau - a_2\xi^2$ has $m = 1$, $dx = 2$ (diffusion), and $\tau^2 - c^2\xi^2$ has $m = 2$, $dx = 2$. A *structure* is then a hypothesis about the shapes of all factors, for example $((1,1),(1,2))$ for "one factor $\tau - a_1\xi$ times one factor $\tau - a_2\xi^2$" (transport times diffusion), whose unknowns are a_1, a_2 and, more generally, all the coefficients allowed by the bounds dx_j.

Explained: a reader asks: exponentials, but exponential polynomials cause ambiguity?

Two different things are easily confused here.

Step 1: what is ambiguous. If two factors share a root, a solution may contain $t\,e^{\lambda t}$ (a polynomial times an exponential), and a mode $e^{\lambda t + i\kappa x}$ lying on two factors' curves can be assigned to either one. The *ambiguity* $\delta(P)$ is therefore about *which factor owns a mode*, not about the representation of the field.

Step 2: why exponentials stay the basis. Every solution is still a superposition of exponentials, with polynomial prefactors only at repeated roots; these are exceptional points (for the wave, only $\kappa = 0$). The representation is complete. Changing to another basis would not remove the ambiguity: it lives in the solution space, not in the choice of functions.

Step 3: what the algorithm does about it. Exponential-polynomial data never identify the annihilator itself, only a law *relative to a class* (theory T13). So (i) a confluent test $(A + Bt)e^{\lambda t}$ labels such cases `UNDECIDED` rather than guessing (Section 4.4); (ii) a shared mode is split equally, the minimum-norm representative (Section 4); (iii) the output never says "identified". Exponentials are the right language; the ambiguity is reported, not hidden.

Why it matters. The unknowns are the coefficients of the polynomials p_k. A mode (κ, λ) is a measurement of the variety: each one gives one equation $Q(\lambda, i\kappa) = 0$ on these coefficients. This is why the first design question has a clean answer inside the linear constant-coefficient class and a hard one outside it (Section 10).

3 Stage 1: interpolation with probabilities

In this section	
Starting from	scattered noisy samples (t_i, x_i, y_i), possibly periodic in x, no knowledge of the equation.
Why this section	the analytic core needs a field on a regular grid, and because data are finite there is no single correct field. The stage must return *many* fields with weights, not one.
Role in the paper	first link of the chain. Its output $\{(\hat{u}_s, \pi_s)\}$ is what the core is run on, sample by sample.

Symbols used here: what they are and where they come from		
(t_i, x_i, y_i), $i \le n$	the n measurements: place (t_i, x_i) and measured value $y_i \approx u(t_i, x_i)$	the data
ε_i, s_n	measurement noise (random error) and its standard deviation	assumed Gaussian
f	the unknown smooth function we fit; it equals the field u up to the constant c	the quantity inferred
c	constant offset (mean level) of the standardised data	preprocessing
kernel $k(\cdot, \cdot)$	a function giving the prior covariance of f between two points: it encodes how smooth f is	the choice of model
$k_t^{(r)}, k_x^{(r)}, R$	factors of the kernel in t and in x for the r-th of R summed terms ("separable" = a product of a function of t and one of x)	the choice of model
s_r, ℓ_t^r, ℓ_x^r	amplitude of term r and its correlation lengths in t and in x (the distance over which f stays correlated)	hyperparameters
η	the list of all hyperparameters $(s_r, \ell_t^r, \ell_x^r, s_n, c)$, in logarithms	unknown, marginalised
$\bar{f}$, K_{post}	posterior mean and covariance of f given the data	output of the GP
N_s samples, $\hat{u}_s$, π_s	N_s fields drawn from the posterior on a regular grid, and their weights $(\pi_s \ge 0, \sum_s \pi_s = 1)$	output of this stage
q, π_h	proposal distribution for η and the importance weight of draw h	Monte Carlo device
H^m	Sobolev space: functions with m square-integrable derivatives; "not continuous in H^m" means a small change in the data's *smoothness-norm* can produce a large change in the output	functional analysis

3.1 The model

A Gaussian process is a probability distribution over functions, specified by a mean and a *kernel* k (a prior covariance between the values of f at two points). In standardised units the data are modelled as $y_i = c + f(t_i, x_i) + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, s_n^2)$ and a Gaussian-process prior

$$f \sim \mathcal{GP}\Big(0,\ \sum_{r=1}^{R} s_r^2\, k_t^{(r)}(t,t')\, k_x^{(r)}(x,x')\Big), \tag{3}$$

a sum of R separable terms. The sum is there for a theoretical reason: $u = \sum_j u_j$ with $\ker P = \sum \ker Q_j$ (theory T1), and the components of different factors have *different correlation lengths* in (t, x) (a travelling bump and a decaying wave do not share one). A single separable term undercovers on such fields. The kernel *family* (`rbf`, `matern52`, `rbf+rbf`, ...) gets a posterior from its evidence.

Explained: what a GP posterior returns, and why samples

Step 1: the posterior is a distribution over functions. Given the data, the posterior of f is Gaussian with a mean $\bar{f}$ and a covariance K_{post}. The mean $\bar{f}$ is a smooth curve through the data. It is tempting to hand $\bar{f}$ to the analytic core, and the pipeline does *not* do this.

Step 2: why not the mean. The mean is smoother than any real realisation (it averages away the fluctuations), so its high-wavenumber content is biased; the core reads exactly that content (modes at large κ). Moreover the maps used downstream are discontinuous: E_{int} is not continuous in H^m (theory T12), and the annihilator depends discontinuously on the data. For such maps "the map of the mean" is not "the mean of the map".

Step 3: sample. Draw N_s functions $\hat{u}_1, \ldots, \hat{u}_{N_s}$ from the posterior (pathwise conditioning, Matheron's rule: an exact prior draw on the grid, corrected by the data). Run the core on each. The spread of the outputs *is* the propagated uncertainty. This is a Monte Carlo estimate, and it is never linearised.

Step 4: the weights π_s. Hyperparameters $\eta = ((\log s_r, \log \ell_t^r, \log \ell_x^r)_r, \log s_n, c)$ are unknown too. Each hyperparameter draw is a Gaussian component of the posterior, and the draws carry weights: $\pi_h \propto p(y, \eta_h)/q(\eta_h)$, where q is a heavy-tailed Laplace approximation (importance reweighting, truncated). So the family $\{(\hat{u}_s, \pi_s)\}$ also carries the uncertainty on the kernel scales.

3.2 Hyperparameters: marginalised, not fixed

Following the design rule that nothing of the loop is set by hand, the hyperparameters are integrated out:

- `laplace`: type-II maximum a posteriori from several starts, a Laplace approximation, then the importance reweighting above. The same weights estimate the evidence of each kernel family.
- `nuts`: Hamiltonian Monte Carlo (NumPyro) for small n.
- `subset` (default for large n): the `laplace` path on a random subset of the data. It is a true posterior given less data, hence conservative (wider) rather than overconfident.
- `ski`: KISS-GP on the output grid, maximum likelihood only. Experimental and *not calibrated*; opt-in.

A deep-ensemble interpolator exists only as an ablation: it is known to be under-dispersed away from the data.

3.3 What has been measured

The coverage of pointwise credible bands was logged on the synthetic cases for $n \in \{400, 2000\}$ and noise 1%, 5%. The nominal-90% band covered between about 0.82 and 0.94 in the logged runs, with an estimated noise level $\hat{\sigma}$ close to the true one. Runtimes in the logs range from about 17 s ($n = 400$) to 80–435 s ($n = 2000$, exact Laplace path). The session report now exists: the done criterion (± 5 points at 50, 90, 95%) is met in 11 of 12 cells with the default Matérn-5/2 family set (the miss is the 50% band at $n = 400$, 1% noise: 0.553, on the conservative side), while an RBF set fails at $n = 2000$ (90% band covering only 0.84–0.86). The estimated noise level is within a few percent of the true one, so the miscalibration, when it appears, is in the function prior, not in the noise. The first end-to-end test (GP samples $\rightarrow$ core) reached only 0.40 weight on the right factors, because the core then took a single scalar white noise; the later per-bin correlated noise in the core (GLS, from the sample draws) removed that obstacle, and Section 14 gives the benchmark that followed. Stage 1 is [implemented, tested].

Why this stage decides the result

The early benchmark on scattered data used a *placeholder* interpolator (one deterministic interpolant, one noise level). There the core recognised the right factor set in 66% (2000 points) and 18% (400 points) of the runs, no better than the weak-form baselines, because the interpolation bias was not propagated. The posterior samples of this stage, together with the refinement on the raw observations (Section 8), are what closed that gap: in the end-to-end benchmark of Section 14 the full pipeline recovers the right set in 0.90–1.00 of runs at 2000 points and 0.73–1.00 at 400. The placeholder numbers are kept here only as the reason the stage matters.

Explained: a reader asks: is the GP fitting the operator, and where is the interaction term?

Step 1: the kernel is not the operator's. The kernel k in (3) is only a prior on *how smooth and how correlated* the field is (correlation lengths ℓ_t, ℓ_x and a scale s). It contains no exponentials, no PDE terms and no dictionary of known terms. Stage 1 knows nothing about P.

Step 2: why a sum of R terms. R counts *correlation scales* (a wide bump and a narrow wave need different lengths), not PDE terms or factors.

Step 3: where is E_{int}? Not here. This stage only returns plausible fields. The interaction term is computed in stage 2 from the coefficients fitted on each sample (Section 4). No dictionary of terms is needed in either stage: the terms are read off the fitted polynomial.

Step 4: the key point. Yes: every value $f(t_i, x_i)$ is modelled as a Gaussian random variable, jointly Gaussian across points with covariance given by k (plus the noise ε_i). The posterior is again Gaussian, but over whole fields, and a sample is one complete field drawn from it.

Why it matters. Stage 1 turns "one imperfect field" into "a weighted cloud of plausible fields". Everything uncertain downstream is obtained by running the unchanged core on each member.

4 Stage 2: the spectral core

In this section	
Starting from	one field $\hat{u}(t, x)$ on a uniform grid (one posterior sample), with a noise level σ; x periodic or windowed.

Why this section	to extract from $\hat{u}$ the factors of the symbol, the component of each factor, its weight and the quantities γ_N, c_F, ρ, together with a posterior probability on the factor structure.
Role in the paper	the only deterministic stage, called as a black box on each sample. It implements the analytic theorems (factor decomposition, interaction term) on numerical data. Two versions exist: v1 (exact data) and v2 (structured fit, used with noise).

Symbols used here: what they are and where they come from

$\hat{u}(t,\kappa)$	Fourier transform of u in x (Sec. 2); on a grid, the FFT	computed from the sample
n_x, dt	number of grid points in x; time step	the grid
y_k, z_m	time series $y_k = \hat{u}(t_k,\kappa)$ at one fixed κ; and $z_m = e^{\lambda_m dt}$	definitions of Sec. 4.1
A_m, λ_m	amplitude and rate of the m-th exponential at that κ	unknowns of the pencil
Y, Y_1, Y_2, V	Hankel matrix of y; Y without last/first column; its leading right singular vectors	linear algebra of the pencil
$\mathrm{snr}_{\min}$, σ, σ_{bin}	minimum signal-to-noise ratio ($= 3$); noise std of the field; noise std of one FFT bin, $\sigma/\sqrt{n_x}$	chosen default; estimated
point $p = (\kappa_p, \lambda_p)$, n	one extracted mode, and the number of them	output of Sec. 4.1
σ_p $(= \sigma_\lambda)$	uncertainty (std) of λ_p, from the Cramér–Rao bound	Eq. (5)
J, g_j	Jacobian matrix of the exponential model and $g_j(t) = e^{\lambda_j t}$	used in Eq. (5)
$d_j(p)$	distance of point p to the curve $Q_j = 0$, in units of σ_p	Thm. 4.1
$\ell(p)$	the label (index of the factor) assigned to p	Thm. 4.1
χ^2, s, n_{par}	robust sum of squared distances; noise scale (≥ 1); number of fitted coefficients	Eq. (6)
BIC	Bayesian information criterion: fit plus a penalty per parameter	statistics
u_j, w_j	component of factor j and its weight	Eqs. in Sec. 4
$a(\xi) = \sum_n c_n \xi^n$	for $m = 1$: $Q = \tau - a(\xi)$, so a is the right-hand side of $u_t = a(\partial_x)u$; c_n are the *term coefficients*	theory (terms of the operator)
a_n, T, $\hat{u}_0$	the n-th term $c_n(i\kappa)^n$; time span of the data; initial profile $\hat{u}(t_0,\kappa)$	definitions
E_{int}, ρ	interaction term and its relative size	Eq. (9)
B_j, S	orthonormal basis of the sampled kernel of Q_j; the matrix $[B_1 \cdots B_N]$	Eq. (10)
γ_N, c_F, $\delta(P)$	separability numbers and the size of the ambiguity	explained in the text

4.1 Modes: FFT in x and a matrix pencil in t

For each wavenumber κ of the FFT, the time series $y_k = \hat{u}(t_k, \kappa)$ is a sum of exponentials,

$$\hat{u}(t, \kappa) \;\approx\; \sum_m A_m(\kappa)\, e^{\lambda_m(\kappa) t}, \tag{4}$$

because each mode $e^{\lambda t + i\kappa x}$ contributes only to the bin κ. The rates $\lambda_m(\kappa)$ are found by the *matrix pencil* (Hua–Sarkar), a linear-algebra form of Prony's method. The result is a cloud of points (κ, λ, A): the data of Section 2.

Explained: the matrix pencil, step by step

Step 1: samples as powers. With dt the time step and $z_m = e^{\lambda_m dt}$, the samples are $y_k = \sum_m A_m z_m^k$. Finding the λ_m means finding the z_m.

Step 2: the Hankel matrix has rank r. Build $Y_{ik} = y_{i+k}$. Each row is a combination of the r geometric sequences $(z_m^{i+k})_k$, so $\operatorname{rank} Y = r$, the number of exponentials. *Example*: $y_k = 3(1/2)^k + (1/3)^k$ gives a Hankel matrix of rank 2 for every size ≥ 2. The order is read from the singular values: those above $\mathrm{tol} \cdot s_{\max}$ are signal, the rest noise.

Step 3: shift invariance. Let Y_1 be Y without its last column and Y_2 without its first. Then $Y_2 = Y_1 \mathcal{Z}$ on the signal subspace, where $\mathcal{Z}$ has eigenvalues z_m. In practice one projects onto the r leading right singular vectors V and takes the eigenvalues of $(V_1)^+ V_2$, with V_1, V_2 the rows of V without the last, resp. first, one.

Step 4: rates and amplitudes. $\lambda_m = \log z_m / dt$. The amplitudes follow from a linear least-squares fit of y on $e^{\lambda_m t_k}$.

Step 5: noise. A bin is used only if its amplitude exceeds $\mathrm{snr}_{\min} = 3$ times the noise level of the bin $(\sigma/\sqrt{n_x})$. In v2 the order is chosen *per bin* by BIC among $r_0, \ldots, r_0 + 2$, where r_0 is the Hankel rank above noise: an exponential merged into another one (unresolved) gets its own pole when the data support it. The pencil error under noise has no theoretical bound (Hankel/Prony conditioning is open, (theory T10)); this is where experiments, not theorems, decide.

4.2 How uncertain is each point

The structured fit of the next subsection compares points to curves *in units of each point's own uncertainty*. For a mode the uncertainty σ_λ is the Cramér–Rao bound of its rate: with $g_j(t) = e^{\lambda_j t}$ and the Jacobian columns $\left(g_j,\ A_j t\, g_j\right)$ for *all* exponentials in the same FFT bin,

$$\mathrm{Cov}(\lambda_j) \;=\; \sigma_{\mathrm{bin}}^2 \left[(J^H J)^{-1}\right]_{\lambda_j \lambda_j}. \tag{5}$$

The bin-wide Fisher matrix matters: a first version that used a 2×2 matrix per exponential underestimated σ_λ when two exponentials interfered, and the fit then overfitted. A floor $s \geq 1$ on a global noise scale (Jeffreys prior) prevents exact data from being overfitted.

4.3 From "fit one P, then factor" to a structured fit

Core v1 fits the lowest-degree monic P vanishing on the points (weighted least squares on the monomials of (2)) and then factors it numerically by tracking the roots $\lambda_r(\xi)$ along a complex path and testing which groups of branches have polynomial elementary symmetric functions. It is exact on exact data and is kept as the test oracle.

It fails under noise. Experiment E1: with relative noise of only 10^{-4}, Klein–Gordon and the telegraph equation were *falsely* factored, transport plus diffusion failed from 10^{-4}, the wave

equation at 10^{-1}. The reason is structural: factoring a polynomial with floating-point coefficients is ill-posed, and no data-free threshold exists (theory T15) (CONV-T5: the discrepancy rule picks the right order, but the threshold cannot be estimated from the data).

Core v2 drops the factoring step. Every candidate *structure* $\mathcal{M} = ((m_1, dx_1), \ldots, (m_k, dx_k))$ is fitted directly on the points, and the structures are compared statistically.

Theorem 4.1 (Structured factor fit: algorithm) (theory T15, T10) [implemented, tested]

Let the points be $p = (\lambda_p, \kappa_p)$ with uncertainties σ_p. For a structure $\mathcal{M}$ with k factors, alternate until the labels stop changing:

1. **assign** each point to the factor curve nearest in uncertainty units,

$$\ell(p) = \arg\min_j \; d_j(p), \qquad d_j(p) = \frac{1}{\sigma_p} \frac{|Q_j(\lambda_p, i\kappa_p)|}{|\partial_\tau Q_j(\lambda_p, i\kappa_p)|};$$

2. **refit** every Q_j by weighted least squares on its own points (the weights are $1/\sigma_p$).

This is "k-means with algebraic curves in place of centres" (a k-varieties regression); 10 initialisations are tried (2 deterministic by sorting branches, 8 random) and at most 30 iterations. The fit is scored by the robust residual

$$\chi^2 = \sum_p \min\left(z_p^2,\ (3s)^2\right), \quad z_p = \min_j d_j(p), \quad \mathrm{BIC}(\mathcal{M}) = 2n \ln\left(\max\left(\tfrac{\chi^2}{2n}, 1\right)\right) + n_{\mathrm{par}} \ln(2n), \tag{6}$$

where n is the number of points, $s^2 = \max\left(1, \mathrm{median}(z^2)/0.455\right)$ is a robust noise scale (0.455 is the median of χ^2_1), and $n_{\mathrm{par}} = \sum_j m_j(dx_j + 1)$ complex coefficients (counted twice unless the field is real). The posterior over structures is

$$p(\mathcal{M} \mid \mathrm{data}) \;\propto\; e^{-\mathrm{BIC}(\mathcal{M})/2}. \tag{7}$$

Intuition

Equation (6) is the profile log-likelihood of Gaussian residuals whose scale s is unknown but cannot be smaller than the Cramér–Rao floor ($s \geq 1$), plus the BIC penalty (a Laplace approximation of the evidence). The cap $(3s)^2$ lets a few points be *outliers* (an unresolved exponential merged into another) instead of corrupting the fit. The candidate set has 85 structures: up to 3 factors, $m_j \in \{1, 2\}$, $\sum m_j \leq 3$, $dx_j \leq 4$, in canonical order so that no structure appears twice.

Explained: the fit on a toy cloud, and what the posterior says

Step 1: the points. For $P = (\tau + \xi)(\tau - 0.3\xi^2)$ and $\kappa = 1, \ldots, 6$ there are two points per κ: $\lambda = -i\kappa$ (transport) and $\lambda = -0.3\kappa^2$ (diffusion).

Step 2: the structure $((1,1),(1,2))$**.** Two factors, each of τ-degree 1, in ξ of degrees 1 and 2. Initialisation by sorting the two branches at each κ by $\mathrm{Im}\,\lambda$ already separates the line (imaginary) from the parabola (real). Refitting each curve by least squares gives $\tau + \xi$ and $\tau - 0.3\xi^2$ up to noise, and $d_j(p)$ is of order 1 for every point.

Step 3: the wrong structures. $((1,2),)$ with a *single* factor $\tau - a_1\xi - a_2\xi^2$ cannot pass through a line and a parabola at once: δ is large for half the points, χ^2 is large, BIC is large. The structure $((1,1),(1,2),(1,2))$ fits equally well but carries more parameters: the BIC penalty makes it less probable. Two structures that differ only by a coefficient equal to 0 (for example $(1,2)$ with $c_2 \approx 0$ versus $(1,1)$) are the same equation, and the posterior splits mass between them: in experiments the top weight ranged from 0.45 to 0.98.

Step 4: reading the answer. A low top weight is *information*, not failure: it signals an ambiguity between structures that the data cannot separate. In the aggregated output (Section 5) the posteriors of all samples are pooled.

4.4 Repeated factors: the confluent test

If the symbol has a factor Q^2, the field contains terms $(A + Bt)e^{\lambda t}$ instead of simple exponentials, and *no finite test can exclude this from finite data* (OPF5B-T6). The core handles it as follows: in each FFT bin with two close poles ($|\Delta\lambda| < 4/T$) it compares, by BIC, two simple exponentials with refined poles (8 real parameters) against one confluent term (6 parameters). If the confluent model wins in any bin the output is labelled UNDECIDED with the reason written out. Before this test both cores returned a confident simple factor for $(\tau + \xi)^2$ (posterior 0.96).

4.5 Components, weights, shared modes

Once the factors are known, each mode is assigned to a factor (the one with the smallest d_j), and the component is the sum of its modes,

$$u_j(t, x) = \sum_{p:\ \ell(p)=j} A_p\, e^{\lambda_p(t-t_0)+i\kappa_p x}.$$

The ambiguity $\delta(P)$. Two factors Q_1, Q_2 may have *common solutions*: functions annihilated by both. If f is one of them, then moving f from u_1 to u_2 (replacing u_1, u_2 by $u_1 + f,\ u_2 - f$) changes neither $Q_1 u_1 = 0$, nor $Q_2 u_2 = 0$, nor the sum u. So the split of u into components is not unique. The theory shows that the freedom is a *finite-dimensional* space of exponential-polynomials, of dimension $\delta(P) = \sum_{i<j} \dim \mathbb{C}[\tau, \xi]/(Q_i, Q_j)$, which counts (with multiplicity) the points (τ, ξ) where Q_i and Q_j vanish together, *i.e.* the modes (κ, λ) that lie on both curves. Example: wave factors $Q_1 = \tau - \xi$, $Q_2 = \tau + \xi$ vanish together only at $(0, 0)$, so $\delta(P) = 1$: the only common solutions are constants. The same-looking number n_{shared} in the code counts such shared modes in the data.

A mode lying on *two* varieties at once ($d < 3$ for both: the intersection responsible for the ambiguity $\delta(P)$ of the theory, (theory T2)) is split equally between them. This equal split is the *minimum-norm representative* of the finite ambiguity and is the posterior mean of the Bayesian layer of Section 7. The weight is

$$w_j = \frac{\|u_j\|_{L^2}}{\sum_k \|u_k\|_{L^2}}. \tag{8}$$

4.6 Terms inside a factor and the interaction ratio

For an evolution factor $\tau - a(\xi)$ with $a(\xi) = \sum_n c_n \xi^n$, the terms are read from the *branch* $\lambda(\kappa) = a(i\kappa) = \sum_n c_n (i\kappa)^n$ by regression on monomials: this is the log-symbol regression of the analytic note, without having to unwrap phases. The many different κ matter: reading terms is a Vandermonde inversion, whose condition number blows up like h^{-d} when the spectral points are clustered at spacing h (theory T9). So the whole band is used, and the number of distinct κ per factor is reported.

With $a_n(\kappa) = c_n(i\kappa)^n$ and T the time span, the interaction ratio is the closed form of the analytic theory,

$$\widehat{E_{\text{int}}} = \Big[\prod_n e^{Ta_n} - 1 - \sum_n \big(e^{Ta_n} - 1\big)\Big]\hat{u}_0, \qquad \rho = \frac{\|E_{\text{int}}\|}{\|u(T) - u_0\|}, \tag{9}$$

at cost $O(N)$ per wavenumber. It vanishes for a single term. Because E_{int} is not continuous in H^m, ρ is used only as a *distribution over posterior samples*; a small positive value for single-term factors in experiments ($\bar{\rho} \approx 0.002$ at noise 10^{-3}, 0.029 at 10^{-2}) comes only from spurious noise-level terms.

4.7 Separability: γ_N and c_F

For each factor take the kernel functions $e^{\lambda_r(\kappa)t + i\kappa x}$ over the band of wavenumbers present (all roots λ_r of $Q_j(\cdot, i\kappa)$), orthonormalise them to a basis B_j of the sampled kernel, and set $S = [B_1 \cdots B_N]$. Then

$$\gamma_N = \min\{\mu \in \text{spec}(S^H S) : \ \mu > 10^{-8}\}, \qquad \gamma_2 = 1 - c_F, \tag{10}$$

and the number of eigenvalues $\le 10^{-8}$ counts the dimension of the shared directions (the $\delta(P)$ ambiguity (theory T2)). This is γ_N of the analytic theory evaluated on the sampled cell, from the full block Gram matrix and *not* from pairwise c_F, which do not determine γ_3 (theory T5). The factor $\gamma_N^{-1/2}$ amplifies every error on the components, and no regularisation can beat it (theory T6): it is an *error floor*, used as an error bar and as a stop criterion. It depends strongly on the cell: for transport plus diffusion $\gamma \approx 5 \cdot 10^{-4}$ for $T = 0.1$, 0.17 for $T = 2$, 0.73 for $T = 8$; it falls about $300\times$ when the spatial window shrinks from 2π to 0.5.

4.8 The output contract

The core never says "identified"

A fit from finite data cannot be certified (no finite certificate exists, (theory T17)). The core returns one of four labels: FIT-COMPATIBLE (every factor has at least $m(dx+1)+1$ modes and at least $dx+1$ distinct κ), ID-KER (the same, but only relative to the declared class, because there are too few modes: a single mode per factor fits infinitely many laws, (theory T13)), MISFIT (more than half of the points are outliers for every structure), and UNDECIDED (no modes above noise, too few modes for any structure, or a confluent mode wins).

4.9 What has been measured

On gridded data with 10 cases (transport, heat, advection–diffusion, wave, beam, Klein–Gordon, telegraph, Airy, Schrödinger, transport plus diffusion), core v2 recognised the right factor set in 58 of 60 (case, noise) combinations up to relative noise 10% in the regression test, and with five seeds per level in the benchmark: 1.00 at noise 0, 10^{-3}, 10^{-2} and 0.90 at $5 \cdot 10^{-2}$. Core v1 reached 0.66, 0.60, 0.56 at the noisy levels. The best sparse-regression baseline (weak-form SINDy) reached 1.00, 0.90, 0.62. The repeated-factor case $(\tau + \xi)^2$ is now reported as UNDECIDED. Open: a mixture outlier likelihood with marginalised outlier fraction (a fixed cap $3s$ is used), transport plus diffusion at noise $\ge 3 \cdot 10^{-2}$, and a false UNDECIDED on the beam at 10^{-1}. The same ten cases tuned the defaults and were then used to measure them; no held-out equations have been tried.

Explained: a reader asks: what is the matrix pencil for, and did we not already build the exponentials?

Step 1: what stage 1 produced. Only numbers: a generic smooth field $\hat{u}_s$ sampled on a grid of (t, x). It contains no exponential model; a GP draw is not written as a sum of exponentials.

Step 2: building the exponential model, here. The FFT in x splits the field into bins, one per wavenumber κ; each bin is a function of t alone. The matrix pencil fits that function as a sum of exponentials $\sum_m A_m e^{\lambda_m t}$ and returns the rates $\lambda_m(\kappa)$. So space is handled by Fourier (modes $e^{i\kappa x}$), time by exponential fitting, and together they give modes $e^{\lambda t + i\kappa x}$: the exponential model in *both* variables.

Step 3: not yet the polynomial. The pencil does *not* estimate the coefficients of the polynomial. It only produces the points (κ, λ). The coefficients of the factors are fitted afterwards, by regression of curves through those points (Section 4.3).

Why it matters. Stage 2 turns each posterior sample into: a posterior over factor structures, the factors with their coefficients, components and weights, γ_N and c_F, ρ, and a label that refuses to claim identification.

5 Stage 3: from many samples to a distribution over factors

In this section

Starting from	for each posterior sample $\hat{u}_s$ a core result: factors $Q_j^{(s)}$ with coefficient vectors $\beta_j^{(s)}$, weights $w_j^{(s)}$, the structure posterior, γ_N, c_F, ρ.
Why this section	the single-sample outputs differ from sample to sample. The stage must merge them into one object that says which factors appear, how often, how strongly, and *how many* there are.
Role in the paper	it produces the cloud $\mathcal{W}$ of the analytic note and the posterior on the number of factors, which later sets the number of groups.

Symbols used here: what they are and where they come from

$\boldsymbol{\beta}_j^{(s)}$	the vector of monic coefficients of factor j in posterior sample s, padded with zeros to the shape $(m \le 2,\ dx \le 4)$	output of the core (Sec. 4)
$w_j^{(s)}$, π_s	weight of factor j in sample s (Eq. (8)); weight of sample s	core; interpolation
$\delta_{\boldsymbol{\beta}}$	"Dirac point mass": one unit of probability placed exactly at the point $\boldsymbol{\beta}$	measure theory
$\mathcal{W}$	the cloud: all the points $\boldsymbol{\beta}_j^{(s)}$, each with mass $\pi_s w_j^{(s)}$	built here
W_j, $\bar{\boldsymbol{\beta}}_j$, Σ_j	total (normalised) mass, mass-weighted mean and covariance matrix of the j-th cluster of the cloud	statistics of $\mathcal{W}$
H, k_{eff}	entropy $H = -\sum_j W_j \ln W_j$ and $k_{\text{eff}} = e^H$: the "effective number" of equally weighted factors	information theory
k	the number of distinct factors (the unknown that the DP posterior estimates)	the unknown
γ_0	concentration parameter of the Dirichlet process: small γ_0 favours few clusters	model hyperparameter
ELBO	evidence lower bound: the quantity a variational method maximises; here used to pick γ_0	variational inference

5.1 The cloud $\mathcal{W}$

Each factor of each sample is encoded as the vector $\boldsymbol{\beta}$ of its monic coefficients, zero-padded to the shape $(m \le 2,\ dx \le 4)$, so that equivalent structures (a $(1,2)$ factor with $c_2 = 0$ and a $(1,1)$ factor) give the same vector. It carries the mass $\pi_s w_j^{(s)}$. The result is the weighted point cloud

$$\mathcal{W} = \sum_{s=1}^{N_s} \pi_s \sum_j w_j^{(s)}\, \delta_{\boldsymbol{\beta}_j^{(s)}}. \tag{11}$$

From it come the dominant factors (modes of the cloud) with total mass W_j, mean $\bar{\boldsymbol{\beta}}_j$ and covariance Σ_j, and the effective number of factors $k_{\text{eff}} = \exp(-\sum_j W_j \log W_j)$. Per-sample distributions of k_{eff}, ρ, c_F and γ_N are kept as *distributions* (mean and quantiles), together with the co-occurrence matrix of factors and the structure posterior pooled over samples.

Explained: why a cloud and not an average

Step 1. Suppose the true law has two factors, transport with $a = -1$ and diffusion with $D = 0.3$. In eight posterior samples the fitted coefficients are -1.00, -0.99, $-1.01, \dots$ and 0.29, $0.31, \dots$, with weights near 0.77 and 0.23. The cloud has two tight clusters, with masses ≈ 0.77 and ≈ 0.23.

Step 2. Averaging the coefficients over samples would be meaningless as soon as one sample returns a different structure (for instance a spurious extra factor). The cloud keeps every outcome with its probability.

Step 3: k_{eff}. For masses $(0.77, 0.23)$, $H = -0.77 \ln 0.77 - 0.23 \ln 0.23 \approx 0.54$, so $k_{\text{eff}} = e^{0.54} \approx 1.72$; a single factor gives $k_{\text{eff}} = 1$ and two equal ones 2. The note then set $k = \lceil k_{\text{eff}} \rceil = 2$.

Step 4: why that rule is fragile. If a noisy sample puts 1.4% of the mass on spurious factors, k_{eff} rises from 2.00 to 2.14 and $\lceil k_{\text{eff}} \rceil = 3$: the true number of laws is 2. A ceiling is discontinuous; this motivates the next subsection.

5.2 A posterior on k: Dirichlet-process mixture

Instead of $k = \lceil k_{\text{eff}} \rceil$, the number of factors gets a posterior. A truncated stick-breaking Dirichlet-process mixture (8 components, diagonal covariances, variational, scikit-learn's Bayesian Gaussian mixture) is fitted on the cloud, separately for each τ-degree m (the degree is an invariant of a factor, so factors of different m never share a cluster). Four details decide whether it works; each was found by a failure.

1. **Tracks are the DP clusters.** Matching factors across samples by nearest coefficient vector, without a distance gate, merged different laws that share m (transport and heat are both $m = 1$): k_{eff} came out as 1.00. A hand-set gate would break the design rule; the DP cluster is the gate.
2. **Every factor once, plus draws.** Replicating each factor 50 times to carry its mass created fake evidence and split heat into 6 clusters. Now every observed factor enters once, plus n extra draws proportional to $\pi_s w_j^{(s)}$.
3. **One isotropic scale.** Standardising each coefficient separately inflated negligible ones (a ξ^3 coefficient of 10^{-4} in some samples and 0 in others became a bimodal unit-scale direction).
4. **A merge step.** With few samples a DP over-splits (Miller–Harrison: the number of clusters of a DP mixture is not a consistent estimator of k). Clusters that *never co-occur in a sample* are merged when BIC prefers one Gaussian with *full* covariance to two. Full covariance matters because fitted coefficients are strongly correlated ($\text{corr}(c_0, c_2) = 0.97$ for heat).

The DP concentration $\gamma_0 \in \{10^{-3}, \ldots, 10\}$ is chosen by the variational lower bound (empirical Bayes). The posterior of k is then a Monte Carlo quantity: draw $s \sim \pi$, a cluster label per factor from the responsibilities, and count distinct labels. A finite-mixture BIC posterior over k is computed alongside.

5.3 What has been measured

On transport plus diffusion (noise 10^{-3} and 10^{-2}, $N_s = 8$): $P(k = 2) = 1.0$; $W = (0.768, 0.232)$ against the true weights; $k_{\text{eff}} = 1.72$; $\bar{\gamma}_N = 0.174$ and $\bar{c}_F = 0.826 = 1 - \bar{\gamma}_N$ as predicted for $N = 2$. On heat: $P(k = 1) = 1.0$ after the merge step. These runs used a *stand-in posterior*: the exact grid plus independent noise draws with $\pi_s = 1/S$, not GP samples.

Thin margins

The merge test for heat at noise 10^{-2} sat at 7.96 against a threshold of 8.32 with $N_s = 8$: it is close to its decision boundary. More samples, or the GP's own covariance of the coefficients, would make it robust. The report recommends the finite-mixture BIC posterior over groups as the group-count rule instead of $\lceil k_{\text{eff}} \rceil$.

Explained: a reader asks: is this where we group, and what is k_{eff} for?

Yes, this stage prepares grouping. It answers *how many* groups to look for (k) and gives each one a centre and a spread. k_{eff} is a first, crude count: the number of factors counted by mass, so that a faint spurious factor adds only a little. Because rounding it up is fragile (Step 4 above), the pipeline uses the DP posterior on k. Note what it does *not* do: it does not group data points and does not find where in (t, x) one law ends and another begins. Here we group *factor estimates* in coefficient space; the grouping of the domain, by local evidence on rectangles, is Section 6.

Why it matters. Stage 3 gives the factor distribution $\mathcal{W}$ and a posterior on how many factors there are. The parameters of the next stage (number of groups, initial centres, variances) are *read from this posterior*, not set by hand.

6 Stage 4: grouping

In this section

Starting from	the distribution $\mathcal{W}$ of factors and the original idea of the project: split the data into groups, each ideally carrying one factor, using an encoder (PCA, SVD, autoencoder) and a clustering algorithm.
Why this section	to decide *what* can be grouped. In a superposition $u = u_1 + u_2$ every data point carries every component, so grouping points cannot separate factors.
Role in the paper	stage 4 replaces "cluster the data points" by clustering in the spaces where factors are actually separate. It is the main conceptual change from the first version of the pipeline.

Symbols used here: what they are and where they come from

cell, window	a rectangle $[t_a, t_b] \times [x_a, x_b]$ of Ω analysed on its own	choice of the algorithm
T, L	time span $t_b - t_a$ and spatial length $x_b - x_a$ of a cell	the cell
n_c	number of grid samples in the cell	the cell
RSS	residual sum of squares: $\|u - (\text{best fit})\|^2$ over the cell	least squares
dof	"degrees of freedom": numerical rank of the matrix of kernel functions $e^{\lambda_r(\kappa)t + i\kappa x}$ used by the fit	linear algebra
n_{par}	number of coefficients of the factors being fitted	the structure
γ_{floor}	smallest allowed γ_N for a cell (0.02 by default)	hyperparameter
$b_1(\Omega)$	first Betti number: the number of "holes" of a region; $b_1 = 0$ for a rectangle	topology
ARI	adjusted Rand index: agreement between two partitions, 1 identical, 0 chance	clustering evaluation

6.1 Why raw data points cannot be grouped

If $u = u_1 + u_2$ with $Q_1 u_1 = 0$ and $Q_2 u_2 = 0$, then at every point (t_i, x_i) the value y_i is the sum of two contributions: no point "belongs" to one factor. The analytic theory says more: no linear encoder on raw fields can make $E_{\text{int}} \to 0$ (theory T8), and sparse or ℓ^1 separation of components acts only on the finite gauge $\delta(P)$ (theory T7). So PCA or an autoencoder on raw fields is excluded, and grouping happens at three levels where the factors *are* separate.

Level	What is grouped	How	Status
A. Mode space	dispersion points $(\lambda, i\kappa)$, by the curve they lie on	k-varieties regression + BIC (Section 4.3)	[implemented, tested]
B. Domain	rectangles of (t, x) that follow different laws	quadtree with evidence, gated by γ_N	[implemented, uncommitted]
D. Hypothesis	the factor structures	posterior $p(\mathcal{M})$, pooled over samples	[implemented, tested]

Table 3: The three levels of grouping (the labels follow the study of grouping types; the "flow family" level C was dropped because the terms already come out of the factor fit).

6.2 Level A: mode space

This is the structured fit of Section 4.3: it separates *superposed* factors. In the (κ, λ) plane a superposition of transport and diffusion is a line and a parabola, which clustering by curve resolves even though the field is mixed everywhere in (t, x). This corresponds to the original idea of grouping so that each group carries one factor, applied where it is possible.

6.3 Level B: the domain

When *different regions* follow *different laws* (for example transport for $x < \pi$ and heat for $x > \pi$), grouping the domain makes sense. Cells are **rectangles**, always: the factor split is proved on simply connected domains and fails on domains with holes (defect up to $\delta(P)\, b_1(\Omega)$, (theory T1)), and rectangles are simply connected by construction. A cell with no holes in a union of groups is checked explicitly.

Explained: the quadtree, step by step

Step 1: evidence of a cell. For a rectangular cell the best structure is scored on the raw samples of the cell by the distance of the data to the band-limited kernel space $\sum_j \ker Q_j$ (variable projection on the factor coefficients). Its evidence is BIC $= \text{RSS}/\sigma^2 + (2\,\text{dof} + 2\,n_{\text{par}}) \ln n_c$. Because σ is known, the BIC of cells covering different samples *adds*.

Step 2: split. Try splitting the cell at fractions of a side. Accept a split if the *sum of the children's BICs is lower than the parent's by more than* 2. At equal evidence ($|\Delta\text{BIC}| < 2$), splits in x are preferred to splits in t.

Step 3: admissibility. A child must still be analysable: $\gamma_N \geq \gamma_{\text{floor}}$ (0.02 by default) and every factor must have at least $dx + 1$ Rayleigh-resolved wavenumbers. This matters because γ_N falls like T^2 and L^2 on small cells: a cell that is too small cannot separate its components and must not be split further.

Step 4: merge. After the recursion, adjacent leaves forming a rectangle with agreeing structure posteriors are merged when the merged evidence is not worse.

Why a special extractor. On a non-periodic small cell, the free two-dimensional mode model is not identifiable: neighbouring κ below the Rayleigh spacing $2\pi/L$ make the Fisher matrix singular. The factors are identifiable though, because u lies in $\sum_j \ker Q_j$; this is why the cell fit works on that kernel span directly. At noise 10^{-3} and 10^{-2} it found the right structure for $L = 2\pi$ and $L = \pi$ where the earlier two-dimensional pencil failed, and on the two-regime field the quadtree split at $x = \pi$.

6.4 Features and clustering of windows

Features are analytic only (theory T8): for each window, the mass of its factors on each *global* cluster of $\mathcal{W}$, the two largest energy fractions and γ_N, each averaged over the posterior samples. The spread of a feature inside a window is used as the covariance floor (empirical Bayes). PCA may be applied on top. Four clustering methods were compared on the rectangles: a Gaussian mixture and k-means (both initialised at the "pure factor" centres $\bar{\boldsymbol{\beta}}_j$ of $\mathcal{W}$, followed by a connected-component split), spectral clustering (affinity masked by adjacency) and agglomerative clustering with connectivity. The number of groups is the MAP of a BIC posterior; the baseline "dominant balance" (GMM on $\log|u_t|, \log|u_x|, \log|u_{xx}|$ per point) was included for comparison.

6.5 What has been measured

With the *exact components* of transport ($x < \pi$) and heat ($x > \pi$) as input, noise 10^{-2}, $N_s = 6$, and windows aligned with $x = \pi$: Adjusted Rand Index 1.0 for all four methods, no group with a hole; dominant balance 0.20–0.35. When one window straddles $x = \pi$ the ARI falls to 0.63 for k-means, spectral and agglomerative clustering, with the number of groups estimated as 3. The structure posterior over k is a poor window feature (heat windows gave $P(k = 1) = 0.72$ on $t \in [0, 1]$ but 0.44 on $[1, 2]$, because mass splits between equivalent structures) and is switched off; the coefficient cloud is the stable carrier of identity.

What this does not show

These grouping results use exact components as input and a boundary that coincides with the window grid. They show that the method works when the regimes are cleanly separated; they do not yet show it on posterior samples from scattered data, nor with a boundary at an arbitrary position.

Why it matters. The idea "one group = one factor" survives in two forms: curves in mode space for superposed factors, rectangles in the domain for regional laws. Raw-point grouping, and encoders on raw fields, are excluded by theorems.

7 Stage 5: the Bayesian output layer (Friedrichs angle)

In this section

Starting from	the factors Q_j returned by the core and the sampled kernel bases B_j on a cell and a wavenumber band; noise level σ^2.
Why this section	the weights w_j are ratios of norms, and how well they are determined depends on how *close* the component spaces are. The layer must turn this geometry into credible intervals and into the probability that the split exists at all.

Role in the paper it uses only the outputs of the core and leaves the core untouched. Its mean coincides with the core's minimum-norm choice, and it adds the spread around it. Status: implemented in `bayes_split.py` (committed in Phase 3), called by the loop on every posterior sample; its coverage is logged but not certified.

Symbols used here: what they are and where they come from

$V_j = \ker Q_j$	the space of functions on the cell annihilated by the factor Q_j (its solutions)	theory
B_j	a matrix whose columns are an orthonormal basis of the sampled V_j	Sec. 4
$S = [B_1 \cdots B_N]$	all bases side by side: the "sum map" that turns coefficients into a field	theory (S of the regularity theorem)
a_j, a	coefficient vectors: $u = \sum_j B_j a_j + \varepsilon$	the unknowns here
ε, σ^2	noise vector and its variance	from stage 1
ω^2	prior variance of the coefficients ($a \sim \mathcal{N}(0, \omega^2 I)$); fixed by empirical Bayes	model hyperparameter
Γ_a	posterior covariance matrix of a	Eq. (12)
θ_k, θ_F	principal angles between V_1 and V_2; θ_F is the smallest nonzero one (Friedrichs angle), $c_F = \cos\theta_F$	geometry of subspaces
M_{split}, M_{single}	the two competing models: two factors versus one	model comparison

7.1 The linear-Gaussian model

Write the field on the cell as $u = \sum_j B_j a_j + \varepsilon$, with B_j an orthonormal basis of $V_j = \ker Q_j$ (sampled on the cell and the band), $S = [B_1 \cdots B_N]$, noise $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$ and a Gaussian prior $a \sim \mathcal{N}(0, \omega^2 I)$.

Theorem 7.1 (Posterior of the components) (theory T5, T6) [implemented, tested]

The posterior of a is Gaussian with covariance

$$\Gamma_a = \sigma^2 \left(S^H S + \frac{\sigma^2}{\omega^2} I \right)^{-1}. \tag{12}$$

For $N = 2$ the eigenvalues of $S^H S$ are $1 \pm \cos\theta_k$, with θ_k the principal angles between V_1 and V_2. In the worst direction the posterior variance is

$$\frac{\sigma^2}{\gamma_2 + \sigma^2/\omega^2}, \qquad \gamma_2 = 1 - \cos\theta_F = 1 - c_F. \tag{13}$$

On directions with $\theta_k = 0$ (the intersection, the $\delta(P)$ ambiguity) the likelihood is flat: the posterior equals the prior, and with an equal-variance prior its mean splits the shared part equally, which is the minimum-norm representative of the core.

Intuition

Equation (13) is the Bayesian form of the analytic bound $\|S^\dagger\| = \gamma_N^{-1/2}$: error bars on the components widen like $\sigma / \sin(\theta_F/2)$ as the angle closes, but are capped by the prior when

$\theta_F \to 0$. In words: the closer two kernels are, the less the data can tell them apart, and the layer says so with wide intervals rather than a false split.

Explained: what the layer outputs, and how

Step 1: weights by sampling. $w_j = \|u_j\| / \sum_k \|u_k\|$ is nonlinear in a, so its interval is computed by Monte Carlo: draw a from the Gaussian posterior and evaluate w_j. No linearisation, in line with the design rule.

Step 2: does the split exist? Compare two models by their closed-form Gaussian evidence: M_{split} ($u \in V_1 \oplus V_2$ with independent priors) against M_{single} (one factor, or the unsplit symbol). With the Woodbury identity, $\log p(u \mid S, \sigma^2, \omega^2) = \log \mathcal{N}(u; 0, \sigma^2 I + \omega^2 S S^H)$ is computed from the small Gram matrix. As $\theta_F \to 0$ the evidence ratio tends to 1 and $P(\text{split})$ returns to its prior: the honest statement that the data cannot separate the factors.

Step 3: the prior scale. ω^2 is set by empirical Bayes: maximise the closed-form evidence on a grid $10^{-6} \ldots 10^{2}$. σ^2 comes from the GP posterior.

Step 4: noise is correlated after interpolation. After GP interpolation the noise has covariance K_{post}, so the angle must be measured in the K_{post}^{-1} inner product (a generalised Friedrichs angle); otherwise the intervals are miscalibrated. In the code this is done as a per-bin generalised least squares: the noise covariance of each Fourier bin is estimated from the interpolation draws (the core's `BinNoise`) and the model of each bin is whitened by it. The full generalised Friedrichs angle in the K_{post}^{-1} product is still approximated bin by bin.

7.2 What has been measured

Experiment E5, with the *true* factors and noise 10^{-2}: $P(\text{split})$ behaves as predicted, and the interval width grows with c_F. The first version had coverage of the nominal 90% interval for w_1 of only 0.65–0.85: undercoverage, attributed to fitting a complex-valued model to a real field. The fixes were made: a real (conjugate-pair) parametrisation (the $\kappa = 0$ bin is real, with a real basis), per-bin correlated noise, and a null test. In the later log (noise 10^{-2}) the coverage of the 90% interval is 0.95–1.00 for transport plus diffusion over $T \in [0.25, 2]$ and 0.85–1.00 for the wave, and on the null (heat with a spurious transport factor) $P(\text{split})$ has mean and maximum 0.000. Coverage above 0.90 on transport plus diffusion means the intervals are now conservative there; the wave at large T is slightly below nominal. The done criterion is coverage within ± 5 points of nominal on transport plus diffusion, wave and beam for $T \in [0.25, 2]$. For $N > 2$ the full spectrum of $S^H S$ is used, not pairwise angles.

Explained: a reader asks: this is a sum, so where is E_{int}?

The sum $u = \sum_j B_j a_j$ is over *factors*. For a linear equation the components of different factors add exactly (superposition), so there is no interaction between them and none is missing. E_{int} lives *inside one factor with several terms*: it measures how far the evolution is from the *sum of the increments* of the separate terms: with $\Delta_n = e^{TA_n} - 1$, $E_{\text{int}} = e^{TA_1} e^{TA_2} - 1 - \Delta_1 - \Delta_2 = \Delta_1 \Delta_2$ (Section 4). It is not an extra component to add to $u = u_1 + u_2$: a factor such as $\partial_t - D\partial_x^2$ has two additive terms but is still one factor, with one component. This layer only asks how well the components can be told apart and whether the split exists; E_{int} is a separate output of the core, reported through ρ.

Why it matters. The layer gives the output a *calibrated humility*: wide intervals and a low split probability exactly where the analytic theory says separation is hard. It is implemented and its logged coverage is close to nominal on the tested cases, but the criterion of ± 5 points on all of them is not yet certified.

8 Stage 6: the loop and the stopping rule

In this section	
Starting from	stages 1–5 for one data set, and the hierarchical idea that the posterior of a parent group is the prior of its children.
Why this section	the pipeline must decide whether the fitted laws are good enough, and if not, what to change: the degree of the factors, the structure, or the grouping of the factors.
Role in the paper	closes the recursion of the original idea. Status: *implemented* in `loop.py` (about 1600 lines, committed in several steps: plan v2.9, then Phase 3). It differs from the first design (kept below as "the original specification") in one respect that matters: it groups *factors*, not regions of the domain.

Symbols used here: what they are and where they come from		
$\mathcal{D}$	the data set: either the gridded field, or the scattered observations (t_i, x_i, y_i)	input
(m, dx) bounds	the largest τ-degree and ξ-degree allowed per factor at this level; start at $(3,4)$, limits $(4,8)$	`START_BOUNDS`, `M_LIMIT`, `DX_LIMIT`
P_{conc}	posterior mass needed to call k or a structure "settled": 0.9	hand default `P_CONCENTRATED`
held-out score	how well factors fitted on the first part of the time window explain the modes of the rest; lower is better	Sec. 8
raw evidence	Gaussian evidence of the *raw observations* inside the span of the factors' kernels (closed form, Sec. 7)	`bayes_split.raw_evidence`
$\bar{\rho}$, $\bar{c}_F$, $\bar{\gamma}_N$	means over posterior samples of ρ, c_F, γ_N	Sec. 4
$P(\text{split})$, $P(k)$	posterior probability of the split, and of the number of factors	Secs. 7, 5

The loop has two halves. This section describes the first: pooling, degree growth, regrouping and recursion, which find a plausible structure. The second half, which refines that structure on the raw observations and labels it honestly, is Section 9; most of the later fixes are there (Section 13).

8.1 What the implemented loop does

Theorem 8.1 (Outer loop, as implemented) (theory T15) [implemented, tested]

Given posterior samples $\{(\hat{u}_s, \pi_s)\}$ of the field (and, when the data are scattered, the raw observations):

1. **Run and pool.** Run the core on every sample and pool the factors into the cloud $\mathcal{W}$ (stage 3). Samples whose factor set is rejected very strongly by the raw observations get no weight.
2. **Degree growth.** If most of the posterior mass is MISFIT or ID-KER, raise the bounds (m, dx) one step. Keep the grown level only if it (a) moves mass to FIT-COMPATIBLE and (b) predicts held-out data better (a guard against fake higher-degree fits).
3. **Regroup at fixed k.** If $P(k)$ is concentrated but the samples disagree on the structure, refit every sample with the consensus structure.

4. **Recursion per factor.** Analyse each factor's component field again on its own; a factor whose component still splits is replaced by its children.
5. **Decide, refine, split.** Choose the structure by the evidence of the raw data; refit the coefficients on the observations; split reducible quadratics and conjugate pairs; run the split layer (stage 5); attach intervals.
6. **Output** the factors with probabilities, intervals and a label. The domain is *never* split by the loop: where each factor carries its energy is only an optional report.

Intuition

A single run asks: given what we can see, is the law *good enough*? If the data are too rich for the allowed degree, grow it, but only if the larger model also predicts data it has not seen. If the samples disagree, force them to agree and look again. If one factor hides two, split it. None of these steps moves the likelihood: they change which candidates are scored, never how they score.

Explained: the pieces the loop calls, one line each

`decide_structure`: among the candidate structures (including sub-structures), choose by the evidence of the raw observations in the kernel span, with a symbol Occam factor, a check that enough distinct wavenumbers exist, and an outlier-bin mixture. This stops the GP's own decay at high κ from being reported as a factor.

`screen_samples`: each sample's factor set is scored by the observations; samples rejected very strongly are dropped from the pool.

`support_path`, `refine_factors` (`refine_raw`): the factor coefficients are refitted to the observations by maximising the closed-form evidence (variable projection), with terms selected by evidence minus an Occam penalty $\frac{1}{2} p \log n$.

`split_quadratics`, `split_branches`, `conjugate_closure`: a quadratic $\tau^2 + p(\xi)\tau + q(\xi)$ is reducible over $\mathbb{C}$ exactly when its discriminant is a square; for a real field the annihilator is closed under complex conjugation, so complex factors are kept with their partner.

`split_layer`: the Friedrichs layer of Section 7 on every sample; `coef_draws` and `bootstrap_intervals`: coefficient uncertainty is propagated into the weights and, for a single noisy grid, a parametric bootstrap gives the intervals.

`domain_report`: a report only, the share of each factor's energy in each block of (t, x).

8.2 The original specification, and what changed

The first design was: stop if the group is clean (at most 2 factors, $\bar{\rho} < \varepsilon_1$, $\bar{c}_F < \varepsilon_2$, $\bar{\gamma}_N$ above its floor, top structure weight ≥ 0.9); otherwise split by evidence (quadtree for regions, mode space for superposed factors) and recurse, with the parent's posterior as the prior of its children. In plan v2.9 this was changed: the loop groups by *factors* only (the domain tree of Section 6 stays a separate tool), because grouping data points cannot separate factors and the observed failures were not grouping failures.

Explained: the loop in plain words

Step 1. Run stages 1–5. **Step 2.** Look at the answer: if the posterior is concentrated on one factor set and one structure, and the held-out check passes, stop. **Step 3.** If most mass is on MISFIT or ID-KER, raise the degree and ask the held-out guard whether the larger level is real. **Step 4.** If k is settled but the samples disagree on the shape, refit all of them with the consensus shape. **Example.** Heat

alone: one factor, no growth (a test checks that the loop does *not* raise the degree). Transport plus diffusion: two factors by the k posterior, grouped by factor. The law $\tau + 0.002\,\xi^5$ needs ξ-degree 5 while the start bound is 4: at the first level no mass is FIT-COMPATIBLE, the bounds grow to $(3, 5)$, the held-out guard accepts the larger level, and the loop returns one factor with coefficient 0.002 (a test checks exactly this).

8.3 Control quantities and who sets them

Quantity	Role in the loop	How it is set	Status
$P_{\text{conc}} = 0.9$	settled k or structure	hand default	[implemented, tested] (hand default)
(m, dx) start and limits	growth path	hand defaults $(3, 4)$, $(4, 8)$	[implemented, tested] (hand default)
γ_N floor	cell admissibility, error bar	fixed at 0.02 from experiments	[implemented, tested] (hand default)
Held-out guard	accept a grown level	prediction of later modes	[implemented, tested]
Top structure weight	structure settled	posterior $p(\mathcal{M})$	[implemented, tested]
$P(\text{split})$	do not split a group with low value	Gaussian evidence (stage 5)	[implemented, tested]
Calibrated thresholds, loss-derived decisions	replace hand defaults	isotonic map and decision rule on a simulator corpus	evaluated, no-go (Section 11); the hand defaults remain

Table 4: Control quantities. "Hand default" means a number fixed from experiments, which the design rule says should eventually be replaced by an inferred value.

What is still not true of the loop

The statement "nothing is set by hand" is still the design intention, not the present state: P_{conc}, the γ_N floor, the signal-to-noise cut, the outlier cap and the degree limits are numerical defaults. A calibration layer meant to replace the decision thresholds was built and evaluated, and found to have nothing to calibrate (Section 11); a controller that would use a diagnosis to choose remedies was not built, because no available trigger finds the failures it could repair.

Why it matters. The loop exists and is tested (Section 14). It is a conservative, evidence-driven procedure whose steps change the candidates and never the likelihood. The part that is still design only is the calibrated stopping rule and the controller above it.

9 Stage 6b: how the final answer is chosen and refined

In this section	
Starting from	the samples of the GP, the factors that the core read from them, and the raw observations (t_i, x_i, y_i) (or the noisy grid).
Why this section	the core reads the factors from an *interpolant*, which is biased exactly where the information is weakest (fast-decaying modes at high wavenumber). The final answer must therefore be decided and refined against the *raw data*, with a fair way to compare models of different size.
Role in the paper	this is the part of the loop where most of the later fixes live (Section 13). It does not change what the earlier stages do; it decides among their candidates and corrects their numbers.

Symbols used here: what they are and where they come from		
B	matrix whose columns are the exponentials $e^{\lambda t + i\kappa x}$ of a candidate's factors, sampled at the data	built from the candidate
τ^2	prior variance of the amplitudes (how large the amplitudes of the modes may be); profiled by empirical Bayes	model hyperparameter
evidence	$\log \mathcal{N}(y; 0,\ \sigma^2 I + \tau^2 \sigma^2 B B^H)$, maximised over τ^2	`refine_raw.log_evidence`
p	number of real coefficients of a candidate (a conjugate pair counts once)	the candidate
Occam term	$\frac{1}{2}\, p \log n$, subtracted from the evidence; n is the number of data	BIC / Laplace
score	evidence $-$ Occam term; the quantity every comparison reads	–
`LOG_BF_BARE`	1 nat: the margin a new model needs to displace the current one	Kass–Raftery scale
support	the set of monomials $\tau^a \xi^b$ that a factor keeps	chosen by the score
excited bin	a Fourier bin κ whose energy exceeds the noise by a χ^2 test	data-only test

9.1 Why the raw data, and what "evidence" means here

The factors found by the core are only a starting point. For a candidate set of factors, the data should be a sum of its exponentials plus noise; *how well* that fits, with the amplitudes integrated out, is a number: the evidence. Because it is a Gaussian integral it has a closed form, so it is cheap and exact. On a grid the same number is computed per pair of Fourier bins (the unitary transform keeps it equal to the dense computation to rounding), which makes it fast.

Explained: evidence and the Occam term, with a small example

Step 1. Take the field $u = e^{-0.3\kappa^2 t} \cos \kappa x$ for $\kappa = 1, 2, 3, 4$ plus noise. Candidate A: one factor $\tau + a\xi^2$, $p = 1$. Candidate B: $\tau + a\xi^2 + b\xi^3 + c\xi^4$, $p = 3$.

Step 2. B always fits the noise a little better, so its evidence is slightly higher. The Occam term $\frac{1}{2} p \log n$ charges each extra coefficient about $\frac{1}{2} \log n$ nats (4.5 nats for $n = 8192$ grid points). B wins only if its fit gain exceeds the charge.

Step 3. A *nat* is the unit of log of a probability ratio; one nat is a factor $e \approx 2.7$. The rule `LOG_BF_BARE` says that a new model must beat the current one by more than 1 nat. Within that margin the model with fewer real parameters is kept: a difference that small is optimiser noise, not information.

9.2 Choosing and refining, in order

1. **Decide the structure** among the candidates by raw evidence (Section 8), with a check that enough distinct wavenumbers are observed: a factor with as many unknowns as equations merely interpolates its modes and is inadmissible.
2. **Refine every scored candidate**, not only the winner: a candidate that was rejected because the core's coefficients were biased can win once refined. Coefficients are fitted to the observations by variable projection.
3. **Select the support.** Starting from the *full* support up to the class degree, terms are removed one at a time along a mode-level least-squares path; each support is scored by evidence minus Occam, and the best is kept. (Two fixes belong here: the path used to start below the true support, and to stop at the factor's own degree, so a term the core had rounded to zero could never return.)
4. **Split.** A factor is split into its branches when the split is the *same kernel*: then no evidence is needed, the split is simply kept, and it is repeated to the finest factorisation. A split that changes the model must beat the unsplit factor by more than the margin. In a real field a complex factor is kept only together with its conjugate.
5. **Drop** factors whose removal raises the score; merge pairs whose product scores higher.
6. **Check the final law against the data** (below) and label it.

Why a model comparison needs one scale per *branch*

The amplitude prior has a scale. If a single scale is shared by all roots in a Fourier bin, a branch that is *not excited* in a bin (for a real pair, the conjugate root in half of the bins) is charged as if it carried signal, and a correct two-factor law can lose to an odd-power approximation with more terms (the case of two conjugate pairs). Giving every excited root branch its own scale removes that charge. It has its own bias: a factor with no excited branch becomes free. The code therefore uses one scale per root branch in the excited bins, pools all empty bins under one shared scale (a test says which bins are empty), and keeps a separate rule that the factor set must pass the goodness-of-fit test on the raw data. This is an example of a fix that moves a bias rather than removing it; Section 13 lists the follow-ups.

9.3 The last line of defence: goodness of fit and honest labels

Theorem 9.1 (Reported law must fit the data) (theory T17) [implemented, tested]

A law that the data reject is never reported as FIT-COMPATIBLE. On scattered data the raw points are projected on the reported law's kernel in the excited band plus two bins; the residual, with the known noise level, is χ^2 with $n -$ rank degrees of freedom. If $\log_{10} p < -3$ the label becomes MISFIT. On a real noisy grid a *smooth* test (the residual projected on a few low-order Legendre polynomials in t, which is a χ^2 with known degrees of freedom) turns FIT-COMPATIBLE into UNDECIDED when $\log_{10} p < -3$. Identifiability (ID-KER) is judged on the *reported* factors, with their own wavenumber support.

Intuition

Two failures look alike from inside the pipeline: the right law is in the class and the search did not reach it, or the data genuinely cannot tell. The goodness-of-fit test catches the first without claiming to cure it: it downgrades the label to UNDECIDED. The identifiability re-check catches the opposite error: a label ID-KER computed on samples of a structure that was later replaced describes a law that is not the reported one.

The false-alarm rate was checked under the true law: on 150 fresh grid draws none fell below 10^{-3}, and on the scattered inputs (with fresh noise, 400 draws) none either.

9.4 Intervals

On a single noisy grid, coefficient intervals come from a parametric bootstrap of the *refined* estimator (each noisy copy is refitted at the final support); on scattered data from a Laplace approximation of the refined evidence. A factor produced by an exact split inherits the interval of the unsplit factor, carried through the split map by Monte Carlo (an earlier version reported zero-width intervals there, which is what made the coverage drop to 0.5). The weights come from re-running the split layer on 16 draws of the factor coefficients.

> **Why it matters.** The loop now has two jobs. The first, of Section 8, is to find a plausible structure; the second, here, is to confront it with the raw observations: refine it, price its terms fairly, split only when it is the same kernel or clearly better, and label it honestly. Most fixes made after the first end-to-end benchmark are here.

10 The two design questions

In this section

Starting from	the two questions put to the machine-learning side: must the basis functions of the interpolation be chosen in advance, and how should the outer structure around the analytic core be designed.
Why this section	to show, in one place, what the algorithm now answers and with what evidence, so that the remaining open parts are visible.
Role in the paper	summary of Sections 3–8 from the point of view of the two questions.

10.1 Q1: must the basis functions be chosen in advance?

Inside the class of linear constant-coefficient equations in $1+1$ dimensions, no. The interpolation is a Gaussian process, which needs no basis, and the "basis" used by the core is the family of exponentials $e^{\lambda t + i\kappa x}$, which is universal in this class (Section 2, (theory T25)). What is chosen in advance is only a *bound*: factors up to degree $m_j \le 2$ in τ (total ≤ 3) and $dx_j \le 4$ in ξ, plus the class itself. A transport equation is found because its curve $\lambda = -i\kappa$ is present among the fitted points, not because a "transport basis function" was in a list.

Where the answer stops

Outside the class (variable coefficients, nonlinear terms) there is no universal basis, and the dictionary problem returns, with the theorem that no countable dictionary identifies every equation exactly and that with a fixed dictionary the equation is recovered only modulo its differential consequences. The core's method depends on constant coefficients: extending

it needs new work (for example frozen-coefficient fits per cell, which the quadtree would support). The kernel family of the GP is also a choice, although its scales are marginalised and the family gets a posterior.

10.2 Q2: the outer structure

Sub-question	Present answer	Evidence	Status
Loop and stopping rule	recurse by Bayes factors; stop when clean with probability > 0.95	tests: heat does not grow, transport plus diffusion is grouped by factor, a fifth-order law grows once under the held-out guard; thresholds are hand defaults	[implemented, tested]
Parameters from $\mathcal{W}$	DP mixture on the cloud gives $P(k)$, centres $\bar{\boldsymbol{\beta}}_j$, covariances Σ_j	$P(k=2)=1$ on transport plus diffusion, $P(k=1)$ on heat; merge step thin	[implemented, uncommitted]
Encoder and clustering	analytic features; GMM, k-means, spectral, agglomerative on rectangles	ARI 1.0 with exact components, aligned windows; 0.63 if straddled; dominant balance 0.2–0.35	[implemented, uncommitted]
Uncertainty through a discontinuous map	Monte Carlo over GP samples; never linearised	GP band coverage within ± 5 points in 11 of 12 cells; the end-to-end benchmark below runs on GP samples	[implemented, tested]
Noise and validation	benchmark against PDE-FIND, weak-form, ensemble and Bayesian SINDy, BOP-DMD	full pipeline, run 6 (30 runs per cell, right factor set at noise $10^{-3}, 10^{-2}, 5\cdot 10^{-2}$): grid 1.00/1.00/0.93, 2000 points 1.00/0.97/0.90, 400 points 1.00/0.93/0.73, at least as good as every baseline in all nine cells. After the later fixes, final benchmark (10 cases $\times$ grid 1%, grid 5%, 400 points 5%): 30/30 right, no false-confident; best baseline (weak SINDy) 23 right, 5 false-confident	[implemented, tested]

Table 5: The second question, point by point.

Takeaway. The point on which the whole proposal depends, "scattered noisy data $\to$ GP samples $\to$ core $\to$ loop", is demonstrated on the ten named cases, which also tuned the defaults. Generality is tested on randomly drawn equations (Section 14): 20 draws, no false-confident answer, but a small sample.

11 Does it need training? The learned layers

In this section	
Starting from	the deterministic core plus the Bayesian layers, which are the reference.
Why this section	to state what "training" means for this algorithm, and how learned components are added without losing the guarantees.
Role in the paper	describes Phase 4: the learned layers were built in separate sessions, evaluated on corpora, and mostly not adopted. The full account of each is in the companion document *The learned layers*.

Symbols used here: what they are and where they come from		
D	a data set	as before
θ	the parameters of the true law: structure and factor coefficients	the target of inference
$q_\phi(\theta \mid D)$	a neural network with weights ϕ that outputs a probability distribution of θ given D	learned
proposal	a candidate (a set of poles, a structure) offered to the exact evidence, which scores it	rule of Phase 4
calibration	an offline map that corrects a probability or an interval width on a declared simulator	rule of Phase 4
flag	a warning attached to a result; it never changes the label	rule of Phase 4

11.1 The reference works without a training set

There is no training data set. The core is analytic; the Bayesian layers do *inference per data set*: the only quantities "learned" are hyperparameters of that data set (kernel scales and noise, the DP concentration, the prior scale ω^2), estimated by type-II maximum likelihood, Laplace approximation, NUTS, or empirical Bayes.

11.2 The one sentence that governs every learned layer

Theorem 11.1 (Rule of Phase 4) [implemented, tested]

A learned component may change *where* the Bayesian algorithm looks and *how fast* it gets there, and it may calibrate a prior on a declared simulator distribution. It may never change *what the data say* about a candidate.

Every learned output is therefore one of three things: a **proposal** (scored afterwards by the exact raw-data evidence, which has its Occam term), a **calibration** (fitted offline, checked by simulation, tagged as prior-dependent), or a **flag**. Each layer has an off switch, and off reproduces today's pipeline bit for bit; the core itself changes only through an optional argument.

11.3 What was built and what was refused

Layer	What it does	Code	Status
L0	learned seeds (pole sets) for the per-bin exponential fit, trained on generic exponential sums	`learned/expsum.py`, `expseeds.py`, hook in the core	mechanism tested; no trained network or benefit result
L2b, L2c, L2d	calibrated $P(\text{top})$; calibrated intervals; learned global constants	`learned/calibrate.py`	**no-go**: nothing to calibrate ($P(\text{top}) \geq 0.999$ on 96% of runs), coverage already 0.895 at 0.90, every fold picks today's constants
H0	deterministic high-κ residual scan	evaluation code	**no-go**: never changed an answer
V0	deterministic battery of diagnostics on a finished run	`validate.py`	[implemented, tested] (used by the loop)
V1	learned failure detector, risk control and an out-of-distribution gate	`learned/validator.py`	**no-ship**: AUROC 0.85 (leave-one-family-out), but detection 0–8% at the required false-alarm level
Controller C'	remedies chosen from a diagnosis	none	not built: oracle ceiling 8/12 failures, but no trigger finds them (0/12)
LB exact memo	caches repeated steps, same output	learned-layers branch	**on**: exact, about 0.5% faster
Simulator	random linear equations by family, truth known by construction	`simulator.py`	[implemented, tested] (the held-out benchmark)
L1b, L6, L5	learned residual proposals; amortised posterior; learned grouping encoder	none	deferred / rejected

Table 6: The learned layers of Phase 4.

Acceptance rule and the risk

A trained component is kept only if, on held-out equation *families*, it gives a right set at least as good as without it, coefficient error and coverage no worse, a false-FIT-COMPATIBLE rate no higher, and passes the null tests (no extra modes on pure noise or on a single-mode bin). The risk is out-of-distribution failure: a network trained on synthetic linear constant-coefficient data says nothing reliable outside that class, so the validator withholds its probabilities when a run looks out of range. The evaluation has now been run: none of the learned layers improved the pipeline, and the one that is on (the memo) changes no output. On the final benchmark "pipeline" and "pipeline plus ML" are the same 30/30. The honest summary is that the deterministic-plus-Bayesian reference, after its fixes, left the learned layers little to correct.

12 A worked example, from data to output

In this section

Starting from	the test case `transport_plus_diffusion`: $P = (\tau + \xi)(\tau - 0.3\xi^2)$, i.e. a superposition of a transport factor $\partial_t + \partial_x$ and a diffusion factor $\partial_t - 0.3\partial_x^2$, with several modes per factor.
Why this section	to follow every stage on one case with the numbers measured in the experiments.
Role in the paper	makes Sections 3–8 concrete. *Honesty note:* the numbers come from a gridded field with relative noise 10^{-3}–10^{-2} and a *stand-in* posterior (the grid plus independent noise draws, $N_s = 8$, $\pi_s = 1/S$); the GP stage was not part of these runs. The end-to-end benchmark of Section 14 does use GP samples.

The field has many modes per factor (it must: with one mode per factor the coefficients are not identifiable, (theory T13)). Transport modes sit on the line $\lambda = -i\kappa$, diffusion modes on the parabola $\lambda = -0.3\kappa^2$.

Step 1 — interpolate

Goal. Obtain plausible fields with weights.
Computation. In the full pipeline: GP posterior from scattered data, S samples $\hat{u}_s$ with weights π_s. In this example: $N_s = 8$ copies of the grid with independent noise, $\pi_s = 1/8$.
Outcome. $\{(\hat{u}_s, \pi_s)\}_{s=1}^{8}$.

Step 2 — modes, per sample

Computation. FFT in x; in each bin with amplitude above $3\times$ the bin noise, matrix pencil in t with the order chosen by BIC; Cramér–Rao σ_λ for each point.
Outcome. Two points per wavenumber: $(\kappa, -i\kappa)$ and $(\kappa, -0.3\kappa^2)$, up to noise.

Step 3 — structured fit, per sample

Computation. The 85 structures are fitted by k-varieties regression and ranked by BIC (6), (7).
Outcome. The top structure is two τ-degree-1 factors, $\tau + \xi$ and $\tau - 0.3\xi^2$ (pooled structure posterior 0.994 on $k = 2$ factors). Components u_1, u_2 and weights w_j follow from (8).

Step 4 — separability

Computation. Orthonormal kernel bases on the cell, $S^H S$, (10).
Outcome. $\bar{\gamma}_N = 0.174$, $\bar{c}_F = 0.826 = 1 - \bar{\gamma}_N$ (as predicted for two factors); each factor is single-term, so $\bar{\rho} = 0.002$ at noise 10^{-3} and 0.029 at 10^{-2}.

Step 5 — aggregate

Computation. Cloud $\mathcal{W}$ (11); DP mixture per m; merge step; $P(k)$.
Outcome. $P(k = 2) = 1.0$; masses $W = (0.768, 0.232)$ against the true weights; $k_{\text{eff}} = 1.72$, so $\lceil k_{\text{eff}} \rceil = 2$.

Step 6 — grouping and loop

Computation. The factors are superposed everywhere in (t, x), so there is nothing to split in the domain (the quadtree finds no split with better evidence); the separation is the one of Step 3 in mode space. The loop's test: ≤ 2 factors, $\bar{\rho}$ small, $\bar{\gamma}_N = 0.174$ above the floor 0.02.
Outcome. Stop. (The implemented loop groups by factor and decides from the posterior on k and the structure; this example was computed with the stage-by-stage scripts, not by a run of `run_loop`, so it shows the decision, not a log of the loop.)

Step 7 — output

Outcome. The equation as a product of factors, $(\partial_t + \partial_x)(\partial_t - 0.3\partial_x^2)$, with credible intervals on the coefficients; the probability of each structure; the weights with intervals (from stage 5, once implemented); label FIT-COMPATIBLE. The output never says "identified".

13 What was fixed, and why

In this section	
Starting from	the algorithm as described in Sections 3–9, and the theory-to-algorithm table of the project (rows T1–T103) in which each change is justified.
Why this section	after the first end-to-end benchmark many failures appeared that no single design decision explained. Each was diagnosed, traced to a cause, fixed, and tested. This section lists them by theme, so that a reader sees the *kind* of problem the algorithm had and what the fix says about how it works.
Role in the paper	this is a guide to the fixes, not a full ledger: each row names its T-number, and the reasoning, the numbers and the test are in the project's table and test files.

Symbols used here: what they are and where they come from

Tn	row n of the theory-to-algorithm table: the justification of a choice	`analysis/THEORY_TO_ALGO.md`
adopted / off	a fix is *adopted* when it is on by default; *off* means the code is kept behind a switch but not used	environment switches
false-confident	the pipeline says FIT-COMPATIBLE with a wrong factor set: the worst outcome	scoring rule
data-equivalent	the reported law and the truth fit the data equally well, up to the noise	Cochran / Wilks test
micro suite	a set of at most 15 inputs aimed at one problem, rerun after each change	`benchmarks/micro.py`
A/B	the same inputs run with and without a change, to see what it breaks	discipline

The working rules behind every fix

Gold rule: a fix must hold for any factor structure, order, coefficient scale and equation in scope, never be tuned to the case that exposed the bug; it is justified from theory or a scale-free criterion, and tested on *held-out* families. **Accuracy before speed:** a speed-up is allowed only if it gives the same decisions. **Keep the structure:** an algorithm inside a step may change, a step changes only if indispensable. **Always A/B:** a change that fixes one input is adopted only if it does not break the suites.

13.1 Seeing the noise correctly

Problem	Fix	Test
GP error is correlated in t, but the core assumed white noise; the first end-to-end run reached only 0.40 weight on the right factors (T37)	per-bin generalised least squares, noise covariance from the sample draws	noise covariance recovers an AR(1) autocovariance; GLS with white draws equals the white core
Complex fields: the noise level was declared inconsistently, so a correct Schrödinger law was reported as ID-KER (T89, T91)	noise of a complex field declared as $\mathcal{CN}(0, \sigma^2)$ (real and imaginary parts each $\sigma^2/2$), on grids and scattered inputs	benchmark input test
A growing mode overflowed $e^{\lambda t}$ and crashed the diagnostic SVD	columns scaled per mode	growing-mode test (fails on the old code)

13.2 Choosing the structure by the data, not the modes

Problem	Fix	Test
Mode-level BIC prefers a fake $\tau + c$ factor at each κ (T39)	decider scores each candidate by the evidence of the data in its kernel span	decider rejects constant factors per κ
A symbol that passes through too few modes merely interpolates them (Schrödinger at 1%, T40, T46)	admissibility counts the *observed* roots per κ, at most m	decider rejects a symbol that interpolates few modes; counts only observed branches
The GP mean adds its own decay as a factor on scattered data (heat at 400 points, T44)	the decider and the factor subset are scored on the raw observations	raw evidence rejects an interpolation artefact
Cubic-in-τ laws were unreachable (T57)	core candidates include irreducible orders up to $m_{\max}$	cubic fit test
Growth was only in ξ (T59)	growth tries $(m, dx + 1)$ and $(m + 1, dx)$	grow options both directions

13.3 Refining numbers on the observations

Problem	Fix	Test
Factors fitted to biased interpolant points carry spurious terms (T51, T52)	coefficients refitted to the raw data by variable projection; supports chosen by evidence minus Occam	refinement tests on grid and dense data
Only the winning structure was refined, so a better one was lost (T56, T72)	every structure the decider scored is refined	Schrödinger spurious $m = 2$ case
A 0.2-nat gain displaced a model through the optimiser's noise (T74, T78)	new model must win by > 1 nat; otherwise fewer real parameters are kept	bare-mention tie rule
The same model under two labels was compared on optimiser noise (T66, T68)	same-model signature on the expanded product	supports identify one model across labels
A term the core rounded to zero could never return; the path started below the full support (T94, T100, T102)	the support path starts from the full support up to the class degree	class-degree path tests; F4-128 case
Grid evidence was slow (T62, T81, T85, T87)	per-bin-pair exact form, vectorised, warm-started, triangular solve (about $100\times$)	equal to the dense value to rounding

13.4 Splitting, and conjugate pairs of real fields

Problem	Fix	Test
A real field with a τ^2 factor needs complex factors (beam, T42, T49)	rerun with complex coefficients; accepted only for a non-real conjugate pair	beam case
A complex split lost its conjugate (T53, T65)	conjugate closure; a split in a real field is rejected unless every non-real factor has its conjugate	conjugate-closure test
Reducible factors of any order, not only square-discriminant quadratics (T54)	branch-partition split, accepted by refined evidence	reducible products split; irreducible heat and transport do not
A split that is the same kernel was decided by noise (T80, T99)	exact split kept without refitting, repeated to the finest factorisation	wave with low κ
Two conjugate pairs: the real fit used odd powers of ξ, and the complex fit needed too many factors (T98)	half-plane fit of a real field plus one amplitude scale per root branch	half-plane test

13.5 Repeated roots, confluence and labels

Problem	Fix	Test
A repeated root looked like a confident simple factor (T60, T67)	confluent test by variable projection and Levenberg–Marquardt; UNDECIDED for a repeated factor	repeated factor is undecided
The branch point of an irreducible law looked confluent (damped beam, T60, T69, T73)	branch-point test; confluent bins re-checked against the final factors and their product	branch point is not confluent
ID-KER computed on samples of a structure later replaced described a law not reported (T103)	identifiability re-run on the reported factors; κ support recomputed on the final factors	final identifiability tests

13.6 Not claiming what the data reject

Problem	Fix	Test
A good fit on the interpolant hid a bad fit on the raw points (T88)	goodness of fit on the raw data for scattered inputs; $\log_{10} p < -3$ becomes MISFIT	calibrated on fresh draws under the true law
Fast exponentials spread over many degrees of freedom escaped the χ^2 test on grids (T96)	smooth test on Legendre polynomials in t; downgrade to UNDECIDED	no false alarm on 150 draws
A subset drop removed the factor that carried the ID-KER label, leaving an untested law (T101)	the goodness-of-fit test runs on the law left after the drop	final goodness-of-fit test
The benchmark counted a data-equivalent law as wrong by summing χ^2 over empty bins (T82, T95)	equivalence judged in the excited band with degrees of freedom adjusted	scoring test

13.7 Intervals and weights

Problem	Fix	Test
A single grid has no posterior spread (T50, T75)	parametric bootstrap of the refined estimator	bootstrap keeps supported terms
Real field fitted with a complex model undercovered the weights (T38)	conjugate-pair parametrisation, empirical-Bayes prior scale per bin	weights cover the truth and the null stays low
Split factors had zero-width intervals (beam, coverage 0.5, T103)	interval of the unsplit factor carried through the split map by Monte Carlo	beam 400 points, coverage 1.0

13.8 Tried and not adopted

Explained: why a fix is sometimes refused

Every candidate fix was run on held-out suites with and without the change. Several were refused, and the reason is as informative as the fixes that stayed.

One scale per bin everywhere (T76). It fixed one case (damped beam) but made empty bins each fit a little noise, and the unsplit wave $\tau^2 - \xi^2$ beat the true split $\tau \pm \xi$ (a regression), and it was 2–4× slower. Replaced by pooling empty bins (T77, adopted) and, later, one scale per root branch (T98).

Growing past the degree limits (T83). Its premise did not hold: the failing case never reached the limit, and with the switch on it still ended UNDECIDED. **A guard on every refinement change, and pre-regroup alternatives (T97).** Each fixed one case but produced false-confident answers on three others, so they stay off with the code kept.

Re-interpolating with the candidate factors' kernels (T84, T90). It rescued a false-confident case but dropped the coverage of the intervals from 1.00 to 0.50 on beam, and cost 2× the time.

Learned layers. The failure detector, the calibration of probabilities and intervals, the global constants and the controller were evaluated on corpora and not shipped (see the companion document on the learned layers).

What the fixes did not close

Open: two conjugate pairs (the true factors are found but the label stays UNDECIDED because of a confluent bin between different pairs); the scoring with one scale per root branch still does not charge a factor with no excited branch (a bare τ); the beam at 400 points still has a 0.5 interval coverage on one seed (an ordinary 90% miss); the weight interval over the ambiguity $\delta(P)$ of a shared $\kappa = 0$ mode; non-monic, nonlinear, variable-coefficient and higher-dimensional laws are outside the algorithm. The cost rose: scattered cases at 400 points take tens of minutes.

Why it matters. Almost every fix is one of three kinds: *see the noise correctly*, *judge candidates by the raw data with a fair price for terms*, or *refuse to claim what the data reject*. None changes the architecture. Together they took the pipeline from 66% and 18% right on scattered data with a placeholder interpolator (an early benchmark) to the right factor set on all 30 inputs of the final benchmark, with no false-confident answer; the two benchmarks differ in design (the final one is smaller and uses one seed), so the comparison shows direction, not a rate.

14 Limits, open problems and status

In this section

Starting from	all earlier sections.
Why this section	to separate in one list what the theory forbids, what is open in the theory, and what is merely not finished in the code.
Role in the paper	honest summary for a reader who needs to decide what can be trusted.

14.1 What the theory forbids

- A fit from finite data cannot be certified; the output is a label and a posterior, never "identified" (theory T17).
- No data-free noise threshold exists (theory T15); thresholds are posterior quantities.
- No method beats the error floor $\gamma_N^{-1/2}\sigma$ on components (theory T6); the expected worst-case noise rate for coefficients is $\varepsilon^{1/3}$ (theory T14).
- E_{int} and the annihilator are discontinuous: uncertainty only by Monte Carlo (theory T12).
- The factor split fails on domains with holes: rectangles only (theory T1).
- The factor split loses one derivative (theory T11): component priors must be one order smoother than the field.

14.2 What is open in the theory

Conditioning of the Hankel/Prony step under noise ((theory T10), open): the practical noise threshold is empirical. A repeated factor has no finite test (it is reported UNDECIDED). The closure question for $(\tau - \xi^2)(\tau + c\xi)$ is open. The algorithm covers only constant coefficients

in $1+1$ dimensions; variable coefficients, nonlinear equations (Burgers, KdV), fractional and Dirac operators and higher dimensions are outside it.

14.3 What is not finished in the code

Item	Status	Note
Core v1, core v2 and its extensions (per-bin noise, higher orders, confluent test)	[implemented, tested]	committed; full suite reported as 238 passed at the last handoff
Aggregation, DP, clustering, domain, baselines	[implemented, tested]	committed in Phase 2
GP interpolation	[implemented, tested]	band coverage within ± 5 points in 11 of 12 cells
Friedrichs layer (`bayes_split`)	[implemented, tested]	logged coverage 0.85–1.00; criterion not certified
Loop and refinement (Sections 8, 9)	[implemented, tested]	committed; thresholds hand defaults (P_{conc}, floor 0.02, degree limits, 1 nat, -3 in $\log_{10} p$)
Scattered data end to end	[implemented, tested]	final benchmark 30/30 right (below)
Held-out random equations	[implemented, tested] (measured)	20 draws: 12 right, 7 equivalent, 1 abstention, no false-confident
Learned layers	evaluated	none ships; an exact memo (no change of output) is on; see the companion document
Controller, L1b, L6, calibration of the hand defaults	[planned]	not built

Table 7: Status of each component at the time of writing.

14.4 The benchmarks, from the first run to the final one

Run 6 (before the later fixes). Ten named cases, 30 runs per cell: right factor set at noise 10^{-3}, 10^{-2}, $5 \cdot 10^{-2}$ on the grid 1.00, 1.00, 0.93; at 2000 scattered points 1.00, 0.97, 0.90; at 400 points 1.00, 0.93, 0.73; at least as good as every sparse-regression baseline in all nine cells; median run time about 5 s.

Final benchmark (P8). Ten cases $\times$ {grid 1%, grid 5%, 400 points 5%}, one seed: the pipeline gives the right factor set on **30/30** inputs, no false-confident answer (run 6 on the same inputs: 26 right, 3 abstentions, 1 false-confident). The sparse-regression baselines on the same inputs: weak SINDy 23 right and 5 false-confident, ensemble SINDy 19 and 8, Bayesian SINDy 20 and 10, PDE-FIND 4 and 25, BOP-DMD 1 and 29. The coverage of the coefficient intervals is 0.94 at nominal 0.90 on all 30 inputs; the weight intervals (only 6 inputs have two factors) cover 1.00. The price: the median run takes about 127 s and the total about 1.9 hours, against 5 s and 26 minutes in run 6, and single scattered cases at 400 points take tens of minutes.

Equations nobody tuned for (E13). Random linear equations from five families (the simulator of Section 11), 20 draws at noise 10^{-2}: 12 right, 7 equivalent (the data cannot tell them from the truth), 1 correct abstention, *no false-confident answer*; before the later fixes: 10 right, 9 equivalent,

1 abstention and 1 false-confident. The sparse-regression baselines get at most 6 right. On the hand-picked held-out suites: 13 of 14 right and 11 of 11 on conjugate pairs. These are small samples: indications, not rates.

14.5 Three cautions on the evidence

First, the ten named cases tuned the defaults and are also the cases that measure them; the random draws above are held-out evidence but few, and the scattered seeds of the benchmark share one simulated field, so repeated misses are not independent evidence. Second, grouping results use exact components and boundaries aligned with windows. Third, the toy example of the first note ($0.3e^{-t}\cos x$) has a single mode for its diffusion factor, for which the coefficient is not identifiable; the examples here use many modes.

> **Takeaway.** The algorithm is solid where its inputs are gridded and its structure is linear, constant-coefficient and $1+1$. The route from scattered noisy data through the GP posterior to the refined, honestly labelled answer works on the named cases and on the first random draws, at a cost in time. The open steps are generality on larger samples, two conjugate pairs (found but labelled UNDECIDED), calibrated weight intervals over the ambiguity of a shared mode, replacing the hand defaults by inferred values, and speed.

15 Glossary of tools

Evidence (marginal likelihood) Probability of the data under a model with its parameters integrated out; higher is better, and it already contains a penalty for flexibility (Sec. 9).

Occam term, nat $\frac{1}{2}p\log n$ charged per model for p real coefficients and n data; a nat is the unit of the logarithm of a probability ratio (1 nat $\approx$ a factor 2.7).

Variable projection Fitting nonlinear parameters (poles) while the linear ones (amplitudes) are solved exactly at each step.

ARD, per-branch scale Automatic relevance determination: one prior scale per component, so an unexcited component costs no prior volume (Sec. 9).

Goodness of fit, $\log_{10} p$ A χ^2 test of the residual against the known noise; $\log_{10} p < -3$ means the law is rejected.

Data-equivalent Two laws whose fits differ by less than the noise allows; the data cannot choose between them.

Matrix pencil Prony-type method: rates of an exponential sum from the shift-invariance of a Hankel matrix (Sec. 4.1).

Dispersion variety $\{P(\lambda,\mu)=0\}$; points $(\lambda, i\kappa)$ are the modes of a solution (Sec. 2).

k-varieties regression k-means with algebraic curves as centres; alternate assignment and least-squares refit (Thm. 4.1).

Cramér–Rao bound Lower bound on the variance of an unbiased estimate; here for the rate of an exponential in a noisy bin (Eq. 5).

BIC / Laplace evidence $-2\ln p(\text{data}\mid\mathcal{M}) \approx \text{BIC}$; posterior $\propto e^{-\text{BIC}/2}$ (Eq. 7).

Confluent mode $(A+Bt)e^{\lambda t}$, produced by a repeated root (Sec. 4.4).

γ_N, Friedrichs angle Stability of splitting a sum into components; $\gamma_2 = 1 - c_F$ (Eq. 10).

Dirichlet-process mixture Mixture model with an unbounded number of components; stick-breaking weights (Sec. 5).

Pathwise conditioning Exact GP posterior samples by correcting a prior draw with the data (Matheron's rule).

Importance reweighting Weights p/q turning draws from a proposal q into a posterior sample.

Adjusted Rand Index Agreement of two partitions, corrected for chance; 1 is identical.

Simulation-based calibration Check that a posterior estimator is calibrated on simulated data (Sec. 11).

16 Primer: the ideas used, at machine-learning level

Why $\log \hat{u}$ is linear in the terms. For a constant-coefficient evolution $u_t = \sum_n A_n u$ each Fourier mode obeys $\hat{u}(t,\kappa) = \hat{u}_0(\kappa) e^{t \sum_n a_n(\kappa)}$, so $\log \hat{u} - \log \hat{u}_0 = t \sum_n a_n(\kappa)$, linear in the unknown term coefficients. A regression on the monomials $(i\kappa)^n$ reads them off, which is the regression of Section 4 on the branch $\lambda(\kappa)$.

Why BIC and not a threshold. A threshold on a residual decides "is P factored?" without a scale. BIC compares models by fit and parameter count, with the noise scale estimated; two models that fit equally well are separated only by the penalty. The theory says no threshold can be right for all data ((theory T15)).

Why Monte Carlo and not error propagation. A first-order propagation $\mathrm{Cov}(\mathrm{out}) \approx J\,\mathrm{Cov}(\mathrm{in})\,J^\top$ requires a differentiable map; the maps here (E_{int}, the annihilator, factor structure) jump. Sampling inputs and recording outputs gives a distribution that can have several modes.

Relation to the original idea. Interpolation with probabilities, an analytic map per sample, a distribution of factors and a recursion on groups are all kept. What changed: groups live in mode space and on rectangles; the number of groups comes from a DP posterior; encoders act on analytic features; every threshold is meant to be inferred.

17 Scope: what this document contains and what it does not

It describes the algorithm as designed in the plan and as found in the code and session reports at the time of writing. It does not reproduce the proofs (they are in the analytic paper and its ledger), does not report new experiments, and does not describe stages marked [planned] beyond their specification.

18 Notes for the author

1. This edition describes the algorithm as it stands at the handoff of 2026-11-01 (main at the T103 commit, local, not pushed). The theory-to-algorithm rows run from T1 to T103; the tags T38–T103 in Section 13 are algorithm choices, mostly empirical, and are not listed in the appendix of theory tags.
2. The loop of Section 8 differs from the first design: it groups factors, not regions of the domain. The original stopping rule on $\bar{c}_F < \varepsilon_2$ is kept as specification only.
3. Numbers in the worked example come from a stand-in posterior on a grid, not from scattered data; the end-to-end figures are in Section 14.
4. Some numerical defaults are hand-set: γ_N floor 0.02, SNR cut 3, outlier cap $3s$, $P_{\text{conc}} = 0.9$, degree limits.
5. The test counts come from the repository's own logs and a collection of the suite (142 tests); the full suite was run on the working tree at the time of writing: 142 passed (19 min 34 s).

19 The theory tags

The tags (theory Tn) refer to rows of `analysis/THEORY_TO_ALGO.md`, which links each choice of the algorithm to a result of the analytic work. "Proved" here means that the analytic work records a complete proof (status *DIMOSTRATO*) or a published theorem with a checked source.

Tag	Result, in one sentence
T1	The solution splits as $\sum_j u_j$, $u_j \in \ker Q_j$, on rectangles and, more generally, simply connected regions; it can fail when the region has holes.
T2	For evolution factors, the size $\delta(P)$ of the ambiguity of the split is the degree of the discriminant; the projector for factor j has poles only where its own branch collides with another.
T5	γ_N (stability of recovering components from their sum) equals $1 - c_F$ for two factors, and must be computed from the full block Gram matrix when $N \geq 3$.
T6	No regularisation, linear or not, recovers components better than the error factor $\gamma_N^{-1/2}$; none removes the interaction term.
T7	Sparse or ℓ^1 separation of components acts only on the finite ambiguity space; the minimum-norm split is exact and simpler.
T8	No stable linear encoder makes the interaction term vanish; terms are read linearly from the *log*-symbol.
T9	Reading terms from spectral points is a Vandermonde inversion whose condition number blows up when the wavenumbers cluster: use a spread band.
T10	**Open**: the conditioning of the Hankel/Prony step (the matrix pencil) under noise.
T11	Splitting loses one derivative: the components are one order rougher than the field.
T12	The interaction term and the annihilator do not depend continuously on the data in Sobolev norms.
T13	Identifiability needs enough distinct modes per factor; data of exponential-polynomial type never identify the annihilator itself, only a law relative to a class.
T14	Under noise ε the worst-case rate for coefficient errors is of order $\varepsilon^{1/3}$.
T15	There is no data-free noise threshold: the right fitting order can be found, but the threshold cannot be estimated from the data alone.
T17	A fit is never certifiable from finite data: outputs are labels (fit-compatible, misfit, id-ker, undecided), never "identified".
T25	Ehrenpreis–Palamodov: every solution on a convex region is a superposition of exponentials on the dispersion variety.